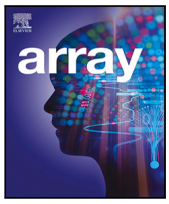
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Multimodal fusion of visual and auditory biomarkers: An epoch-wise stochastic modality masking framework for stroke detection using a CNN–GRU network

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ABSTRACT

The automated identification of neurovascular impairment, particularly ischemic or hemorrhagic stroke, remains challenging due to the multimodal nature of symptoms such as facial asymmetry and dysarthria. This study introduces a multimodal diagnostic framework that integrates visual and auditory streams for accurate and interpretable stroke detection, with a focus on efficient utilization of available data. On the visual side, facial landmarks are derived from MediaPipe's Face Mesh to identify unilateral muscular deficiencies. For the auditory streams, speech recordings are transformed into Mel-Frequency Cepstral Coefficients (MFCC). Notably, the facial and audio data come from two separate sources and lack patient-level correspondence. A primary innovation is the incorporation of an epoch-wise stochastic modality masking approach within the training loop, utilizing a CNN–GRU scheme. In each epoch, a new batch of training data is created by dynamically combining samples from three different categories: complete multimodal input (both facial landmarks and audio MFCC), only facial landmarks and only audio MFCC. This compels the model to learn both unimodal and cross-modal representations, demonstrating the ability to leverage multiple modalities even when true multimodal correspondence across datasets is not available. The empirical results show that the developed framework works well, achieving 95.33% accuracy in multimodal settings. While the unimodal baselines (facial-only: 86.87%; audio-only: 97.41%) show strong performance individually, the proposed modality masking approach enables the model to learn robust representations from cross-source data, demonstrating the potential for stroke detection in preliminary evaluation scenarios.

1. Introduction

Stroke has remained an urgent global health issue of the 21st century, it is the second most common cause of death in the world and the most common reason of long-term disability in adults [1]. For those who survive stroke, many are left with permanent cognitive, motor and additional psychosocial issues that compromise quality of life. Strokes are usually defined broadly as two types: ischemic, which occurs when the vessels supplying blood to the cerebral area of the brain become occluded, leading to death of brain cells, and hemorrhagic, which occurs from the rupture of blood vessels, causing rapid tissue damage through both mechanical injury and the inflammatory cascades [2]. The implications of both ischemic and hemorrhagic stroke are grave, it is worthwhile noting that ischemic stroke represents almost 85% of all strokes globally [3]. An important first step in alleviating the burden of stroke is accurate diagnosis at the earliest possible stage, ideally within an accessible healthcare setting. The headline data tell a sobering story: between 1990 and 2019, the global incidence rate of stroke has rose by more than 70%, mortality increased by 43%, and disability-adjusted life years (DALYs) surged by 143% [3,4]. Among the major challenges facing global healthcare is the growing rate of stroke related morbidity and mortality toward lower and middle-income countries (LMICs), where infrastructure for advanced imaging, such as computed tomography (CT) or magnetic resonance imaging (MRI), as well as specialized clinical personnel, are often absent [1,3]. Given these trends it is obvious that the accurate detection of stroke not only represents a clinical obligation but also constitutes a growing global public health concern.

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In stroke care, timing has a direct correlation with clinical outcomes, with the first hour, often termed as the 'golden hour', being the most critical period for treatment. During this timeframe, clot-dissolving medication (thrombolysis) or clot-removal procedures (thrombectomy) offer the highest chances of successful recovery [5], whereas after this period, the chances of absolute neurocellular death and the patient having serious long-term disability increases drastically. Conventional imaging modalities, like CT, MRI and angiography are very effective in determining the correct stroke subtype [1], nevertheless, they are limited in time to be used as diagnostic approaches because they are expensive, require additional infrastructure along with being time sensitive to obtain, especially in a pre-hospital or resource poor environments [6]. Emergency medical services (EMS) invariably use abbreviated clinical assessments (which can often translate into the pre-hospital EMS systems) for recognition of stroke such as the Cincinnati Prehospital Stroke Scale (CPSS) and the Face-Arm-Speech Test (FAST) [7–9]. These systems rely heavily on observable clinical features such as facial drooping, arm, weakness and slurred speech, all of which have been particularly valuable in triage settings. They are, therefore, limited as they rely on subjective human judgement, do not always agree between observers, and they commonly miss atypical stroke features [7]. Hence, imaging and clinical screening features play a indispensable role in stroke evaluation. These limitations restrict the widespread use of such diagnostic approaches, especially in LMICs, where inadequate infrastructure prevails, making reliable pre-hospital identification essential.

Recent developments in Artificial Intelligence (AI) and Machine Learning (ML) represent a possible solution to such limitations, by enabling automation and objectifying stroke detection through the analysis of multimodal patient data [10]. This is particularly advantageous since ML systems excel at recognizing latent patterns that arise from heterogeneous signals, and are well-suited for tackling clinical problems where symptoms are multi-factorial and subtle. The initial stages of research have explored using unimodal approaches. With respect to visual data, recent studies have employed convolutional neural networks (CNNs) for detecting facial asymmetries, ptosis, and aberrant oral movements, achieving classification accuracies between 90% and 95% on curated datasets [11–27]. With respect to auditory data, models which rely on the Mel-Frequency Cepstral Coefficients (MFCC), and features associated with jitter, shimmer, and prosody have achieved classifications accuracies around 98% in detecting dysarthria, a known speech or motor speech disorder that is commonly associated with stroke [28–33].

The shortcomings of unimodal systems are a recurring theme throughout the studies. Models based on visual data can be highly sensitive to imaging conditions including surrounding lighting, pose and background [17,18]. Similarly, speech-based models may struggle from environmental noise contamination, cross-linguistic effects, and clinical situations where it is not possible to elicit speech (e.g., aphasia; intubation) [32,33]. As well as, from a standpoint of clinical service deployment, it is not practical to use unimodal approaches in this manner, making them brittle, unreliable, and not suited for making acute timely decisions.

These limitations have spurred investigation into multimodal machine learning, which sorts complementary signals in order to improve robustness, accuracy and clinical relevance [10]. The integration of facial and auditory biomarkers align well with established clinical scales, such as FAST and CPSS, that prioritize both motor and phonetic impairments. By combining modalities, multimodal systems leverage the advantages of each modality while offsetting their respective weaknesses. Preliminary studies in this area have demonstrated promising results with multimodal fusion frameworks, indicating greater accuracies than the unimodal baselines [34–40]. Yet, several major problems still persist. First, the scarcity of large-scale, clinically collected multimodal datasets, where visual and auditory signals are collected from the same patient cohort. Second, none of the studies on architectures directly addresses the issue of missing or corrupted modalities, which are common occurrences in clinical practice when visual or auditory data is not available or noisy [38,40]. Finally, prior work has focused squarely on accuracy metrics and relatively little on the far greater issue of interpretability, an absolute requirement for clinical implementation, as the clinicians must trust and understand the algorithm's output [10]. Together these limitations point to a clear research need for the development of multimodal stroke detection frameworks that can effectively leverage multimodal data originating from different sources, are robust to the absence of modalities and are interpretable enough for adoption into real-world clinical workflows.

The study aims to address these obstacles by creating a multimodal diagnostic approach that combines using facial landmarks and speech-derived features (MFCC) in a CNN-GRU model. On the visual side, the geometric asymmetries associated with stroke-related motor impairment can be captured by facial landmarks produced by MediaPipe Face Mesh [41]. On the auditory side, the MFCC features show spectral–temporal properties of dysarthric speech [28]. The components of both modalities are fused using a late-integration approach that provides the advantage of modality-specific feature learning and cross-modal complementarity.

A significant aspect of this study is the introduction of an epoch-wise stochastic modality masking technique, where the two distinct datasets, containing facial landmarks and audio MFCC, are first combined. Then, for each training epoch, the data is dynamically partitioned into three sections: one with both modalities, one with audio masked (retaining only facial landmarks) and one with landmarks masked (keeping only audio MFCC). Unlike typical multimodal model training, this approach compels the model to learn both unimodal and multimodal representations, making it more robust to missing or degraded input, a reality of real-world clinical and pre-hospital assessments.

Explainable AI (XAI) methods using different machine learning algorithms are implemented to rank importance of the features, which allows the XAI solution to focus on the most diagnostic facial landmarks and acoustic features of the speech, generating better generalization and interpretability, since clinicians can trace algorithmic predictions to physiologically meaningful biomarkers.

This study makes several contributions. It shows multimodal fusion outperforms unimodal systems while tolerating modality dropout, incorporates stochastic masking that enables effective learning from cross-source heterogeneous data and improves robustness to data loss, and highlights the importance of interpretability as a key factor in bridging the gap between AI and clinical practice.

The main contributions of the study can be summarized as follows:

- An XAI-enabled feature selection pipeline is developed to identify diagnostically relevant attributes, filtering discriminative facial landmarks and MFCC features. This design enhances interpretability by linking model outputs to physiological measures.
- A CNN–GRU multimodal architecture is designed with dual channels for spatial facial features and temporal audio sequences, fused through dense layers. The framework balances spatial accuracy and temporal sensitivity while remaining scalable to broader healthcare tasks.
- An epoch-wise stochastic modality mixing strategy is introduced that can leverage facial and auditory modalities originating from different sources to train a multimodal model. This approach allows the model to learn cross-modal representations, even when true multimodal correspondence across datasets is not present.
- A systematic evaluation is conducted across unimodal, adapted unimodal, and multimodal settings. Results demonstrated the additive benefits of multimodal fusion and provide a functional basis for clinical deployment.

The rest of this paper is structured as follows. Section 2 reviews related work and highlights the current state of stroke detection using machine learning. Section 3 presents the proposed methodology,

including data acquisition, preprocessing, feature extraction and selection with clinical interpretation, model design, hyperparameter tuning, model evaluation, training strategy and the performance metrics used. Section 4 reports the experimental results, including comparative analysis across unimodal and multimodal conditions, as well as model robustness under degraded input conditions. Finally, Section 5 concludes the study with key takeaways, limitations and future directions.

2. Literature review

Researchers have explored different machine learning approaches for detecting stroke at an earlier stage with an aim to improve patient outcomes, rehabilitation prospects, and long-term quality of life. Early studies mainly focused on visible indicators of stroke such as facial asymmetries for diagnosis. Later on, those ideas were extended into facial image and video-based approaches, allowing researchers to better identify the subtle patterns associated with stroke. In parallel, audio-based detection techniques leveraged changes in speech and articulation as indicators of stroke. More recently, multimodal approaches integrated both visual and auditory cues, offering a more robust framework for the detection of stroke. Table 1 summarizes the most relevant existing research on diagnosing stroke using various machine learning strategies.

2.1. Early studies using static facial images

Among the earliest developments in automated stroke detection in facial image-based systems are the studies that identify facial asymmetries, which in most cases serve as the most pronounced signs of stroke. Initial research in this field is based on 2D fixed facial images which though limited in scope, formed the basis for future advancements. One study [11] evaluates 12 CNN-based facial asymmetry stroke detection models, using deep CNN architectures such as Xception, ResNet50, VGG19, and EfficientNetB1. Although the findings are encouraging, with the best performing model achieving accuracy and AUC of 0.82 and 0.91, respectively, the study has many limitations, including the lack of diverse data, absence of real-life clinical validation, and the exclusion of 3D-data due to noise. Similarly, another research [12] focuses on developing a machine learning system to detect stroke by analyzing facial symmetry. The system utilizes Ensemble of Regression Trees (ERT) to localize key landmarks on the face such as slope, area ratios, structural similarity, Local Ternary Pattern (LTP) and Gabor filters. Feature selection is refined using a method named RelieF-based Genetic Search with Forward Feature Selection (RGSFFS). The study employs Support Vector Machines (SVM) and Bayesian classifiers, both of which achieve a perfect accuracy score of 100% on their relatively small dataset of only 69 images (35 stroke cases, 34 non-stroke), while the Random Forest (RF) classifier reaches 95.45%. Even though the results are robust, the sample size is small and cannot be generalized. Nonetheless, the experiment is limited due to lack of data availability, and the poor generalizability of the models with regard to new cases, which is typical in early phase research. Despite these limitations, early studies provide a foundational understanding of how facial images can be utilized to diagnose stroke, highlighting the potential of non-invasive diagnostic approaches.

2.2. Advances in facial image-based systems

Additional research focuses on developing more resilient and error-free systems relying on facial images. A more advanced pipeline that includes the deep features extracted from Inception-V3 and EfficientNet-B0 along with a hybrid CNN-BiLSTM model is proposed at [13]. This solution, optimized using Multi-Objective Sailfish Optimization (MOSFO), achieves an impressive precision of 98.60%. It outperforms traditional models such as CNN-LSTM, SVM, and models based on Inception and ResNet V2 in both accuracy and inference time. Still, similar to previous models, it relies on curated datasets without external validation. A similar study [14] proposes an application for real-time stroke detection called "StrokeChange", which applies EfficientDet for asymmetry quantification. "StrokeChange" tests three models: activation-only, activation-classification, and activation-regression, with the regression model outperforming others (MAE: 1.255), but the model still suffers from overfitting problems due to subjective manual severity scores. Another research [15] introduces a real-time stroke detection system, "TeleStroke", which employs facial images and federated learning on Jetson edge devices. Using various versions of YOLOv8 (n, s, m, l, x), it leverages multi-scale, anchor-free detection and trains the model using 3745 images (1245 stroke, 2500 non-stroke). Additionally, federated training using FedAvg is done to ensure decentralized learning. An alternative approach uses 2D grayscale facial images for diagnosing stroke [16]. The article's comparison of CNN, RNN, LSTM and GRU on a database of 1663 images shows CNN to be the most effective, achieving 94.29% test accuracy and 97.15% system accuracy. However, this research is limited by the lack of demographic data and sample size, limiting clinical applicability Another notable effort [17] utilizes facial landmarks and contour detection with MediaPipe BlazeFace and FaceMesh, extracting 468 facial key points and 68 critical areas to measure asymmetry around eyes, mouth and jaw. Though it is effective, the approach relies heavily on non-diverse data and variations in image quality, as well as the orientation, constraining performance. In another study [18] on automatic facial landmarks localization for facial palsy evaluation show poorer model performance when applied to patient data compared to healthy subjects. A CNN based computer system [19] for non-contact at-home facial palsy detection is trained on 570 images (200 diagnosed cases) using OpenCV, Dlip, and Keras, demonstrating high accuracy of 98% on collected data and 99% on patient images. But it shows sensitivity to external factors such as patient positioning and movement, limiting the performance in real-time cases, especially in distinguishing actual from simulated symptoms. [42] proposed the Reference Heatmap Transformer (RHT), which introduces reference heatmap information to enhance landmark detection under heavy occlusions and large pose variations. Their approach consists of a Soft Transformation Module (STM) and a Hard Transformation Module (HTM) that cooperate to accurately transform reference heatmap information and model facial shape constraints. Building on similar principles, [43] developed the Multi-order High-precision Hourglass Network (MHHN) to achieve robust facial landmark detection under extreme poses and heavy occlusions. Their method introduces a heatmap subpixel regression (HSR) approach that combines subpixel detection loss and subpixel detection technology, along with a multi-order cross geometry-aware (MCG) model that enhances feature representations by introducing multi-order cross information. While these landmark detection methods focus on accuracy and robustness, [44] proposes a self-supervised learning method (HNCL) for facial expression recognition that does not rely on labeled datasets. Instead of just using augmented versions of the same image, it finds similar expressions through heatmap analysis to create better training samples. This approach captures more natural variations within expression categories and achieves competitive results with supervised methods.

In a broader perspective, a large-scale review tracing the progression of the models used to detect facial palsy phenomenon starts with the classics of mathematical modeling and moves to the use of modern approaches of deep learning. It emphasizes the application of 2D/3D facial asymmetry index and effectiveness of deploying deep learning models compared to the conventional machine learning model, particularly with transfer learning and data augmentation. Although the results are promising, several concerns still remain such as overfitting, the dependency of manual annotation, the heavy job of computer resources, and grievous data shortages and imbalance [20].

Table 1
Summary of the most relevant existing research on stroke diagnosis using machine learning techniques.

Reference	Dataset	Methodology	Results	Limitations
[11]	Training: 185 stroke patients, 551 matched controls; Test: 38 stroke, 59 controls	Ensemble of four CNNs (Xception, ResNet50, VGG19, EfficientNetB1) fine-tuned with age/sex matching; 3:1 control-to-stroke ratio	AUC: 0.91, Acc: 0.86	Single hospital dataset; limited generalizability; no real-world integration; 3D data not utilized
[12]	69 facial images (34 non-stroke, 35 stroke)	Five modules: face detection, landmark localization, feature extraction (LTP, Gabor), structural similarity, RGSFFS feature selection; classifiers: SVM, RF, Bayes	SVM: 100%, RF: 95.45%, Bayes: 100%	Very small dataset; sensitivity, specificity, AUC not reported; risk of overfitting
[13]	Kaggle dataset: 2400 images (1200 stroke, 1200 non-stroke)	ENDDFTL-ABSPFI hybrid CNN–BiLSTM model with MOSFO optimization	Acc: 98.60% (80/20 split), 97.09% (70/30 split); 9.07s compute time	Single Kaggle dataset limits generalizability
[16]	FER-2013 Happy class (650) + 1013 stroke images (web)	Comparison of CNN, RNN, LSTM, GRU on grayscale, downsampled, augmented data	CNN: Train Acc 100%, Test Acc 94.29%	Small, non-clinical dataset
[36]	253 participants (132 stroke, 121 control)	Multi-modal: ResNet-50 (video) + VGGish CNN (audio) + AMMA fusion	Acc: 82.6%	Small dataset; complex model; noisy audio; not mobile-friendly
[37]	Synthetic: 888 videos, 90 audio clips mimicking stroke	MediaPipe, OpenPose, HOG, Eigenface + AdaBoost, SVM, CNN, DNN, ResNet	Acc: up to 100% (some tasks); 47% (leg paresis)	Synthetic data; biased samples; no external validation; high complexity
[38]	221 patients (159 stroke) with NIHSS-based video/audio tasks	Video: ResNet-34; Audio: ResNet-18; adversarial fusion; FairFace/ImageNet pretraining	Acc: 72.40%; Sens: 88.05%; Spec: 32.26%; AUC: 0.7163	Small sample; 3+ min runtime unsuitable for real-time
[39]	Mixed datasets: TORGO (150 audio), NCBI/NIH (700 vascular), NIH eyeGENE (200 retinal), facial landmarks	3-tier: CNN–RNN (audio), SVM (vascular), CNN (retina), AAM+SVM (facial); final SVM fusion for risk score	Overall Acc: 95%	Small datasets; normalization challenges (lighting, noise, face angle)

2.3. Video-based approaches for dynamic facial analysis

Facial image systems have potential, but they lack dynamic analysis of features that are essential for the accurate diagnosis of stroke. Video based systems are influenced by the need to capture temporal data, which may give a better insight on facial movements unlike static facial images, capturing movements during speech or in an expression, offering a much more in-depth view.

One of the studies [21] in this area employs landmark-oriented facial symmetry features extracted from video frames for classifying the severity of facial palsy. The methodology attains more than 95% classification accuracy, but the dataset imbalance is an issue since it constrains the system reliability and generalizability. A different approach uses the Kinect V2 sensor to track the dynamic markers of the Facial Animation Units (FAUs), encoding the movements of the facial muscles [22,23]. The system records an accuracy of 96.8% in accessing the severity of facial palsy. Yet, the data used in this study is mainly focused on synthetic or healthy control subjects, raising questions concerning the applicability of the system in real world scenarios, particularly among stroke patients with heterogeneous and complicated symptoms. In another study [24], a dynamic landmarks search is integrated into the "auto-eFACE", which features a user-friendly interface. The system demonstrates better consistency as compared to manual clinical scoring, suggesting that video based systems may actually offer better consistency than traditional approaches. But like most video based systems, auto-eFACE is limited by its inability to detect subtle, nuanced facial movements, which is a crucial aspect for early stage diagnosis of stroke. Another research [25] presents a user controlled annotation model of gesture-based stroke detection, combining hand and face tracking with feature extraction (position, orientation, speed) and machine learning. The system has an accuracy score of 88.06% using the Lucas–Kanade algorithm for corner tracking and a user-initialized skin color detection model, significantly outperforming a random baseline. The system still demonstrates fairly poor values of precision (47.24%), recall (34.41%) and F1 score (38.71%), implying room for improvement. Its weaknesses primarily include user-dependency, subject to skin color, proximity to the background, and the quality of video. A comparative study [26] examines the effectiveness of machine learning algorithms versus paramedics in detecting facial dysmorphia. Their findings highlight that machine learning models can outperform paramedics in certain situations. However, the algorithm still faces certain challenges due to real-world environmental factors such as lighting conditions, camera angles and occlusion, limiting the adoption of video based systems in the field. Another similar study [27] designs a facial expression recognition model for assisting in monitoring the recovery of stroke patients, especially for those with communication disabilities. The model uses facial landmarks, such as interpalpebral fissure and lip corners, along with decision tree classifiers. It achieves an accuracy of 95%, with precision and recall rates of 91% and 100%, respectively, yet, considering a small number of features may not be the ideal approach for detecting subtle stroke indicators, even in mild cases. Also, the use of decision trees and actor-based data limits the performance of the model in real-world, uncontrolled environments.

2.4. Audio-based approaches for dysarthria detection

Dysarthia is one of the common symptoms of stroke, manifesting as slurred or imperfect speech. Audio-based diagnostic systems present a complementary solution to a non-invasive scalable and efficient modality of assessing the speech impairment of stroke patients. An audio-based system based on deep learning CNN-GRU models, trained on features such as MFCC and short-time Fourier transform (STFT) is utilized to identify dysarthric speech [30]. The system achieves an accuracy of 98.38%. Yet, similar to earlier research, the limitations are the use of small dataset and the absence of external validation. Another research [31] trains a CNN-SVM classifier on the TORGO speech database and obtains a classification accuracy of more than 98%, compared to CNN (93.87%) and DNN-LSTM (96.18%). This work highlights the effectiveness of deep features combined with conventional machine learning classifiers in speech-based detection of stroke.

A dysarthria classifier [32] trained with MFCC, CQCC, speech-specific, and i-vector features, achieves 93.97% accuracy (speaker-dependent) with iMFCC+DNN on TORGO and UA-Speech datasets. But in spite of that, the performance drops to 49.22% on the speaker-independent task, indicating overfitting. Limited generalizability arises from the small dataset size, training–testing class imbalance, and absence of multimodal fusion, while i-vectors remain effective in detecting severity. A classification model [33] of dysarthria severity levels, developed using MFCC, Jitter, Shimmer, HNR, and prosodic characteristics, shows 77.28% and 79.17% accuracy scores with SVM and RF, respectively, moreover up to 99.2% with CNN and LSTM on TORGO and UA-Speech. While the model demonstrates superiority over conventional methods, its performance is constrained when identifying mild dysarthria and subtle differences. In addition, stroke patients do not always exhibit speech abnormalities.

2.5. Multimodal fusion approaches

Multimodal systems that use different types of data, including heart rate, blood pressure, ECG, EMG, video and speech, create a more complete and context-sensitive environment for diagnosing stroke. One framework [34] combines CNN-based image features with health records, where VG16 is used for the images and a 1D CNN for structured data, achieving 98.53% accuracy and 0.99 AUC. A recent study [35] introduces a two-stream stroke rehabilitation monitoring system, consisting of GRUs and CNNs, utilizing facial video and body motion. The system reaches nearly 100% accuracy but suffers due to a small dataset and low variance in patient actions, limiting generalization. Researchers [36] also apply deep learning with FAST and CPSS stroke criteria to analyze speech and facial expressions in 253 patients. The model utilizes ResNet50 for video, VGGish for audio and a fusion strategy called Asymmetrical Multi-Modal Attention (AMMA), achieving 82.6% accuracy and 0.882 AUC. Nonetheless, it struggles with noisy audio and heavy processing loads, causing difficulties in mobile deployment. Some studies also create digital versions of NIHSS scores using synthetic symptoms in healthy people, where HOG, MediaPipe and OpenPose features are combined with SVM and AdaBoost classifiers for detecting dysarthria and arm weakness with 100% accuracy [37]. Still, as it relies on simulated data, clinical conclusions cannot be reached. The ‘DeepStroke’ [38] system uses a two-branch network for video and audio input, applying adversarial learning methods to detect strokes in short recordings of ER patients. The system is tested on 221 patients, proving to be 7%–11% more sensitive and achieves an AUC of 0.7163. Another mobile application, ‘M3Stroke’ [40], targets mild and moderate stroke detection through the analysis of speech and facial expressions. It combines features from ResNet (visual), Audio Spectogram Transformer (audio) and a state space mode (speech sequences). The study promotes fairness by separating demographic factors like race and ethnicity. It achieves an overall accuracy of 80.85% on 269 patients, outperforming traditional triage in subtle cases, but still depends on a limited dataset. ‘StrokeSave’ [39] is another system, where multiple modalities such as vitals, facial video analysis, speech and retina images are combined for early detection of stroke. By using CNN-RNN, AAM and SVM models, the system achieves an overall accuracy of 95% in 327 test cases, outperforming clinical exams. Even so, the strength of the system depends heavily on data quality, as well as quantity, so it also needs stronger validation.

Overall, multimodal stroke detection systems outperform older unimodal methods and move beyond simple static face analysis to real-time video and speech. These systems are better at handling noise and occlusion, but still face technical challenges such as synchronization, dataset limitations and edge device deployment. Despite these obstacles, such systems show great promise for accurate and clinically viable stroke detection.

3. Methodology

This section outline the research methodology of the study. Fig. 1 illustrates the complete process for diagnosing stroke using a multimodal model that combines both facial landmarks and audio MFCC. Initially, a series of preprocessing techniques are applied on the gathered datasets, including removing low quality images and audio samples. Following this step, facial landmarks are extracted from facial images, while audio MFCC are derived from audio recordings. Feature selection strategies are then applied to identify clinically relevant facial regions and the most informative MFCC for the prognosis of stroke. The following sections offer a detailed description of the proposed hybrid CNN-GRU based multimodal architecture, which has two distinct branches, each designed to process one modality of input, along with the hyperparameter tuning approach. Finally, a detailed description is provided of the model training and evaluation strategy, as well as the performance metrics used to judge the overall effectiveness of the proposed model.

3.1. Data acquisition

This section summarizes the procedures and resources applied to ensure accurate, high-quality data collection.

3.1.1. Facial images dataset

The facial image dataset is obtained from a study conducted to diagnose stroke based on facial features [41]. It includes 1000 RGB images, collected from publicly available platforms such as Kaggle and Google Images. Half of these images (500) feature individuals who are clinically diagnosed with stroke, while the other half represents people without any visible stroke symptoms. Special care is taken to maintain diversity in age, gender and ethnicity, ensuring inclusivity in the dataset.

To improve the dataset’s variety and help the model generalize better, several data augmentation techniques such as image flipping, scaling and resizing are applied. In particular, horizontal flipping is performed using the $Transpose$ function from PILLOW library. This step doubled the number of images, reaching a total of 2000 samples (1000 original and 1000 augmented). To confirm the dataset’s demographic balance, DeepFace is utilized, which is a powerful pre-trained deep learning model for recognition and analysis of facial features. This tool accurately predicted key facial attributes such as age group (classified as young adults, middle-aged, or senior citizens), gender and ethnicity by analyzing the facial landmarks.

Only high-quality images, having good lighting and clear facial landmarks, are considered for the final dataset. Furthermore, significant emphasis is placed on the facial images of elderly individuals, as they are more likely to suffer from stroke.

3.1.2. Audio recordings dataset

The audio recordings used in this study are collected from a publicly available dataset on Kaggle [45]. The recordings are sourced from the TORGO database [46], a well-established collection developed through a collaboration between the University of Toronto’s Department of Computer Science, the Department of Speech-Language Pathology, and Holland-Bloorview Kids Rehabilitation Hospital. It includes both acoustic and articulatory recordings from 15 speakers: 8 dysarthric speakers (5 males, 3 females) and 7 control speakers (4 males, 3 females). A speech-language pathologist evaluated each participant’s dysarthria severity, classifying it as severe, moderate, or mild using the Frenchay Dysarthria Assessment. Speeches are recorded over multiple sessions, with each speaker contributing to a minimum of three hours of speech. These recordings included single words, entire sentences, and descriptions of photographs. Single words are selected from a diverse set of sources such as English digits, international radio alphabets, and phonetic contrast words. Meanwhile, sentences are drawn from the TIMIT database and Yorkston–Beukelman intelligibility

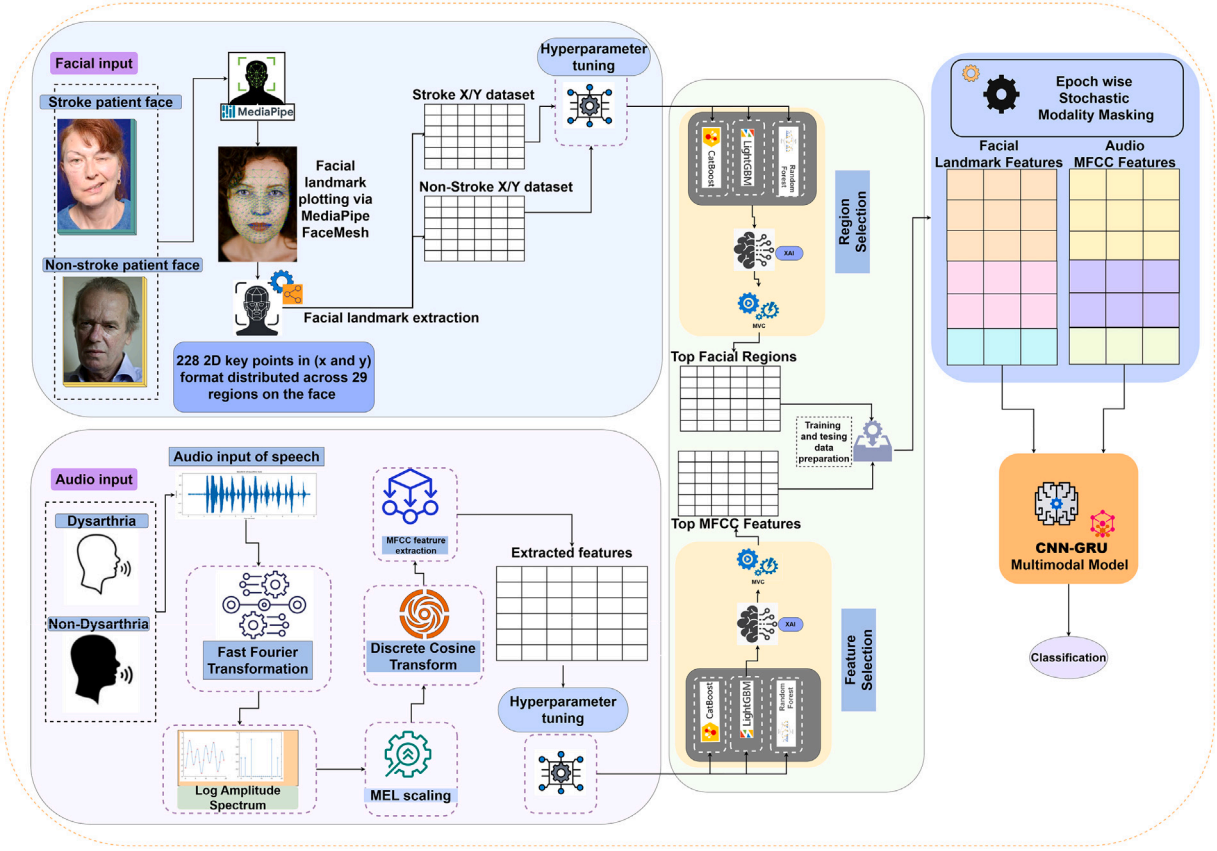


Fig. 1. Proposed multimodal architecture combining facial and audio features for stroke detection.

assessments, both standard sources for speech research. Participants are also requested to describe photographs in their own words, thereby encouraging a more natural, impromptu form of speech, better reflecting real-world speech patterns. Both head-mounted and directional microphones are used to capture high-quality audio. In addition, articulatory data are collected using the AG500 electromagnetic articulography (EMA) system, which tracks the precise movements of the tongue and other speech organs.

For this study, a subset of 1996 audio samples is utilized, which is divided into four categories: dysarthric males (500), dysarthric females (482), non-dysarthric (control) males (500), and non-dysarthric (control) females (514). The distribution is chosen to maintain a near-balanced representation across both conditions and genders.

Each audio recording is linked to a session-based speaker identifier that follows the TORGO naming convention. Speaker IDs begin with M for male speaker and F for female speaker, followed by the session number (e.g. M01, F03). For control (non-dysarthric) speakers, the letter C is appended after the gender identifier (e.g. MC01). This labeling scheme clearly distinguishes dysarthric and control speakers, as well as individual speakers within the same category, a critical aspect for preventing speaker leakage during training and testing.

3.2. Data preprocessing

The section describes the data preprocessing techniques, including data cleaning, extraction of facial landmarks from facial images and MFCC from audio recordings, followed by data normalization for ensuring consistent scaling.

3.2.1. Data cleaning process for facial images

The dataset contains 228 2D (x and y) key points, representing facial regions commonly associated with stroke symptoms, including the Lips (43 points), Eyes (114 points), Eyebrows (29 points), Nose (4 points) and Cheeks (38 points). The dataset goes through a thorough screening of where low-quality and misaligned images are removed, and the remaining pictures are resized and cropped, ensuring the model solely focuses on the facial region.

3.2.2. Extraction and normalization of facial landmarks

The facial landmarks from the images are extracted using MediaPipe Face Mesh, which can extract a total of 468 unique facial landmarks. However, since not all landmarks are clinically relevant for stroke diagnosis, 10 silhouette landmarks are excluded, and consequently, the remaining 458 points are utilized to create 228 2D key points in (y/x) format corresponding to 29 anatomically relevant regions, as shown in Fig. 2. Specifically, 43 landmarks are selected from the lips (11 from the upper outer contour, 10 from the lower outer contour, 11 from the upper inner contour, and 11 from the lower inner contour). Meanwhile, the right and left eyes are represented by 57 landmarks each, consisting of three upper eyelid layers (seven landmarks each) and four lower eyelid layers (nine landmarks each). In addition, the eyebrows consist of a total of 28 landmarks, with 8 upper and 6 lower points selected for both the right and left eyebrow. Furthermore, 5 landmarks are extracted from the nose region, including the midpoint between the eyes, nose tip, nose bottom, right and left nose corners. Finally, 19 landmarks are extracted from each side of the cheeks, summing up to 38 key points.

Subsequently, to enhance model robustness, the landmarks are normalized by considering the nose tip as the primary reference point and subtracting the coordinates of all other landmarks from the position of the nose tip, as shown in Eq. (1), thereby effectively making the nose tip the origin [41].

$$\mathbf{L}'_i = \mathbf{L}_i - \mathbf{L}_{\text{nose}}, \quad i = 1, 2, \ldots, 468 \tag{1}$$

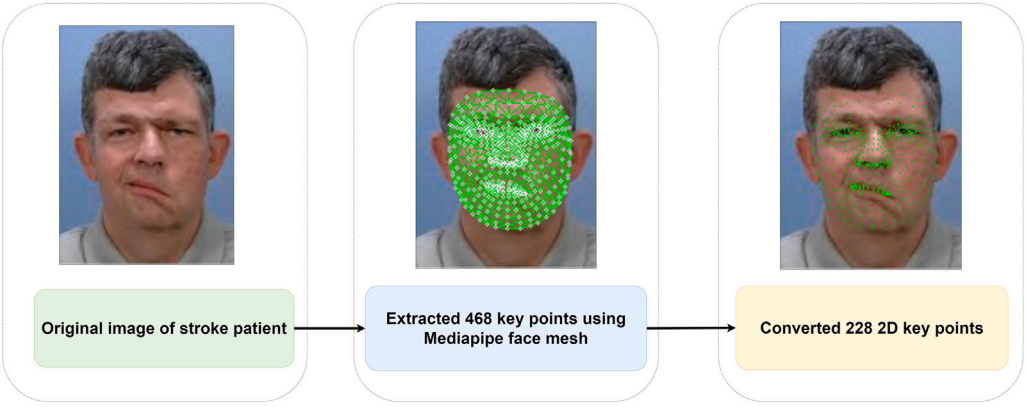


Fig. 2. Extraction of 228 2D key points using MediaPipe.

here $\mathbf{L}_i = (x_i, y_i)$ represents the original landmark coordinates, and $\mathbf{L}_{\text{nose}} = (x_{\text{nose}}, y_{\text{nose}})$ is the nose tip landmark. Following this, to make features face size agnostic, the inter-ocular distance is used as a scale factor (s), which is calculated using Eq. (2).

$$s = \sqrt{(x_{\text{RE}} - x_{\text{LE}})^2 + (y_{\text{RE}} - y_{\text{LE}})^2} \tag{2}$$

In Eq. (2), the points $(x_{\text{RE}}, y_{\text{RE}})$ and $(x_{\text{LE}}, y_{\text{LE}})$ denote the coordinates of the right and left outer eye corners, respectively.

3.2.3. Data cleaning process for audio recordings

The audio dataset is carefully examined to ensure all files are present, intact and free from corruption. This systematic process made sure all the recordings are in the correct format, such as WAV and MP3, ensuring that the sampling rates are consistent while the audio durations fell within the expected range. Furthermore, all files are verified to have the correct audio channels, either mono or stereo. These extensive reviews ensured the dataset is clean, consistent and ready to be used for further processing.

3.2.4. Extraction of MFCC features from raw audio signals

The raw audio signals are converted into Mel-frequency Cepstral Coefficients (MFCC), which is a feature representation technique used in audio signal processing such as audio recognition and audio classification. MFCC represents the spectral characteristics of audio recordings, transforming complex raw signals into a format suitable for machine learning models. At its core, MFCC captures the essence of the power spectrum of a sound signal, making it easier for models to interpret and analyze human speech. Fig. 3 illustrates the process of extracting MFCC features from audio signals.

The speech recordings go through a series of transformations in order to be converted into a usable format. Initially, the sounds undergo Fast Fourier Transform (FFT), which converts the time-domain signal into the frequency-domain, allowing analysis of how different frequencies change over time. Once FFT is applied, the resulting signal is transformed into a log amplitude spectrum. By applying a logarithmic scale, the most significant frequencies are emphasized while suppressing the less relevant, low-amplitude frequencies. After this, the data undergo Mel scaling, a technique that adjusts the frequencies to align with how humans perceive sound. The human ear is more sensitive to lower frequencies [47], especially in speech, and less sensitive to higher frequencies. This scaling mimics that by compressing higher frequencies and focusing more on the lower ones. The frequency spectrum is divided into Mel bins, which represent ranges of frequencies spaced according to the Mel scale, making lower frequencies closer together and higher frequencies spaced farther apart. Once the Mel frequency bins are computed, the Discrete Cosine transform (DCT) is applied, which reduces the dimensionality of the Mel bins by eliminating redundant or less informative data, preserving only the important components that contribute the most in determining the characteristics of the audio signal. After applying the DCT, this transformation yields the final MFCC features.

In this work, the $feature.MFCC()$ function from the Python *librosa* library is utilized to generate a total of 128 MFCC features from each signal, capturing different aspects such as spectral shape, harmonic structure, pitch, and energy distribution. To simplify the feature set, computation of the mean of the MFCC across time frames is done. As the audio signal is time-varying, computing the average of MFCC over the given period helps to capture the overall spectral characteristics of the entire signal, reducing the dimensionality and variability caused by short-term fluctuations [48].

3.2.5. Machine learning algorithms

This section describes the six distinct machine learning algorithms utilized in this study.

Random Forest (RF): RF is an ensemble technique, where multiple decision trees are built and their results are aggregated to give the final output. Each tree gets trained on a bootstrapped sample of the data (random sampling with replacement). After training, the model makes predictions by averaging the outputs of individual trees (for regression) or by taking the majority vote (for classification).

Extra trees classifier (ET): ET is an ensemble learning method that randomly constructs multiple decision trees, similar to an RF algorithm. Contrarily, instead of choosing the best split at each node for an individual tree like RF, ET chooses a random subset of features at each split and selects the best split among these randomly chosen features. Multiple trees are trained on different subsets of the data, and their predictions are averaged or voted in the case of classification tasks to make the final decision.

CatBoost (CB): The CB algorithm is a gradient-boosting strategy leveraging decision trees and gradient boosting methods to improve model performance. At its core, CB utilizes the concept of ordered boosting, where trees are built by treating the target values as categorical, reducing the risk of overfitting while increasing the accuracy of the predictions. The algorithm computes the gradients of the loss function at each step and then updates the model's predictions accordingly. With each iteration of boosting, a weak learner in the form of a decision tree is added to the model, specifically designed to minimize the residual error and refine model performance. The update rule for CB is shown in the Eq. (3).

$$F_t(x) = F_{t-1}(x) + \eta \cdot \nabla L(F_{t-1}(x), y) \tag{3}$$

Where $F_t(x)$ is the prediction after t iterations, $F_{t-1}(x)$ is the prediction from the previous iteration, η is the learning rate, ∇L represents the gradient of the loss function with respect to the model's predictions, and y is the true label.

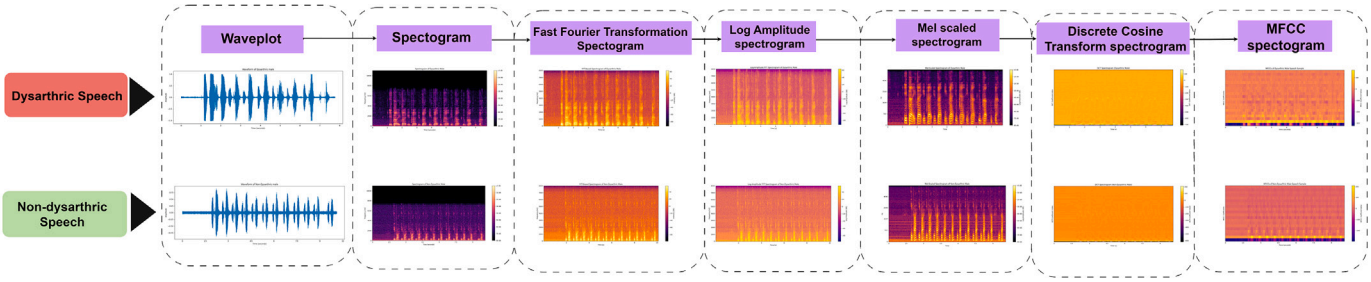


Fig. 3. Extraction pipeline of MFCC features from dysarthric and non-dysarthric speech signals.

Light gradient boosting machine (LGBM): LightGBM is an ensemble gradient boosting framework designed for efficiency in terms of memory usage and computational speed compared to other gradient boosting techniques. The algorithm uses a histogram-based technique for splitting continuous features, reducing memory consumption and improving computational speed. It also incorporated exclusive feature binding and a leaf-wise tree growth strategy, making it faster than traditional gradient boosting approaches. The update rule is shown in Eq. (4).

$$F_t(x) = F_{t-1}(x) + \eta \cdot h_t(x) \tag{4}$$

Here, $F_t(x)$ is the model's prediction after t iterations, $F_{t-1}(x)$ is the prediction at the previous step, $h_t(x)$ is the weak model (tree) added at iteration t, and η is the learning rate.

Extreme gradient boosting (XGBoost): XGBoost is an advanced boosting algorithm that optimizes decision trees using a gradient descent approach. It builds the trees in a sequential manner, with each new tree learning from the errors (residuals) of the previous trees. The algorithm incorporates several regularization strategies such as L1 and L2 regularization to avoid overfitting issues. XGBoost's optimization process aims to reduce both bias and variance by fitting trees to the residuals of previous iterations. The update rule of XGBoost is similar to Eq. (4).

Gated recurrent unit (GRU): A GRU is a type of recurrent neural network (RNN) designed to address the vanishing gradients issue, which traditionally hampers RNNs' long-term dependency learning capabilities within sequential data. A GRU architecture consists of two primary gates: the update gate and the reset gate, responsible for regulating the flow of information across the network. The update gate controls how much of the previous hidden state (h_{t-1}) is carried forward to the current time step and how much of the new candidate hidden state ($\tilde{h}_t$) gets incorporated. On the other hand, the reset gate (r_t) handles the extent to which the previous hidden state is being discarded during the calculation of the candidate hidden state. Eqs. (5), (6), (7), and (8) are responsible for governing the operation of GRU.

Update gate:

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t] + b_z) \tag{5}$$

Here, σ represents the sigmoid activation function, and W_z and b_z are the weight matrix and bias term associated with the update gate, respectively.

Reset gate:

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t] + b_r) \tag{6}$$

In this case, W_r and b_r are the weight matrix and bias term corresponding to the reset gate.

Candidate hidden state:

$$\tilde{h}_t = \tanh(W_h \cdot [r_t * h_{t-1}, x_t] + b_h) \tag{7}$$

The candidate hidden state is an intermediate calculation that reflects what the model proposes as the new information to be added to the current hidden state. W_h and b_h represent the weight matrix and bias term in this calculation.

Final hidden state:

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t \tag{8}$$

The final hidden state, h_t, is a weighted combination of the previous hidden state, h_{t-1}, and the candidate hidden state, $\tilde{h}_t$, with the weights governed by the update gate, z_t.

In these equations, h_t represents the hidden state at time step t, while x_t denotes the input at time t. The weights W_z, W_r, W_h and biases b_z, b_r, b_h are learned during training.

3.3. Feature selection

Initially, 228 key facial landmarks are extracted from the dataset, corresponding to 29 clinically relevant facial regions. Each landmark is represented by normalized (x, y) coordinates to account for facial variations. Furthermore, the dimensionality is reduced by computing the gradient (y/x) for each landmark, transforming the 2D coordinates into a single 1D value that captures the proportional vertical to horizontal displacement, a crucial indicator of stroke-relevant facial asymmetry. Following this step, feature importance is calculated by applying three distinct machine learning models: RF, XGBoost, and CB. Each model is trained and evaluated over five randomized train–test splits. After training, the model assigns feature importance scores for each landmark. Subsequently, the region-level importance is calculated by summing the importance scores of all the landmarks within a particular region. This process of aggregation is repeated for each of the five train–test splits, and the resulting region-level importance scores are summed across iterations, producing the final cumulative importance of each region for each distinct classifier.

As shown in Fig. 4, after computing the importance of each region for all classifiers, the final score of a region is obtained by summing the scores across the three classifiers for that specific region. Based on these scores, the regions are ranked and the top 10 regions are selected for further analysis.

Three distinct classifiers are utilized to find the most relevant MFCC features: CB, LGBM and ET. Each model is trained on the given dataset and outputs a feature importance score. Since the raw importance scores can vary across models, they are normalized using Eq. (9).

$$\tilde{I}_j(f_k) = \frac{I_j(f_k)}{\max_m I_j(f_m)}, \quad j \in \{\text{CB}, \text{LGBM}, \text{ET}\} \tag{9}$$

here $\tilde{I}_j(f_k)$ represents the normalized importance score of feature f_k from model j, $I_j(f_k)$ the raw importance score of feature f_k from model j and $\max_m I_j(f_m)$ the maximum importance score across all features for model j After this, the normalized importance scores are aggregated by computing the average importance of each feature, as shown in Eq. (10).

$$\bar{I}(f_k) = \frac{1}{3}\left(\tilde{I}_{CB}(f_k) + \tilde{I}_{LGBM}(f_k) + \tilde{I}_{ET}(f_k)\right) \tag{10}$$

where $\bar{I}(f_k)$ represents the average importance score of feature f_k $\tilde{I}_{CB}(f_k)$ the normalized importance score of feature f_k from CB,

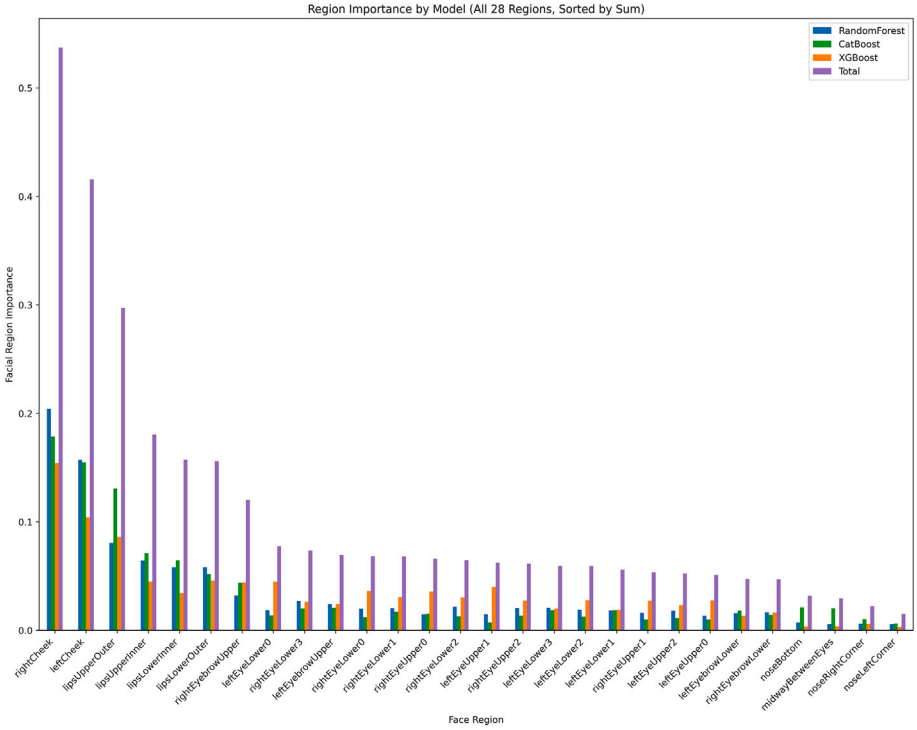


Fig. 4. Feature importance of facial landmark regions across RF, CB and XGBoost models.

$\tilde{I}_{LGBM}(f_k)$ the normalized importance score of feature f_k from LightGBM and $\tilde{I}_{ET}(f_k)$ = normalized importance score of feature f_k from ET.

Finally, based on their aggregated scores, the features are ranked in descending order, and the top 116 features are selected for subsequent evaluation.

As mentioned earlier, for the facial landmarks, the top 10 regions are identified, which include right cheek (19 landmarks), left cheek (19 landmarks), upper outer lips (11 landmarks), upper inner lips (11 landmarks), lower inner lips (11 landmarks), lower outer lips (11 landmarks), right upper eyebrow (eight landmarks), right lower eye (nine landmarks), left lower eye (nine landmarks), and left upper eyebrow (nine landmarks), resulting in a total of 116 landmarks. For the audio features, the pipeline discards the least informative 12 MFCC, specifically MFCC such as 125, 110, 73, 4, 122, 105,127,74,23,17, 121, and 75. To ensure consistency between the facial and audio modalities, the top 116 MFCC features are selected, ensuring that both modalities provide the same number of input features and that the dimensionality remains consistent during model training.

3.4. Clinical interpretability of feature selection

The feature selection process for the facial landmarks identifies top 10 facial regions corresponding to the cheeks (right cheek, left cheek), lips (upper outer lips, upper inner lips, lower inner lips, lower outer lips), eyebrows (right upper eyebrow, left upper eyebrow) and lower eye (right lower eye, left lower eye) regions, as shown in Fig. 5. These selected regions demonstrate strong alignment with established clinical stroke assessment protocols, particularly the Face-Arm-Speech-Time (FAST) protocol widely used by emergency medical services. The FAST protocol specifically instructs clinicians to observe facial asymmetry by asking patients to smile and checking for uneven facial movement precisely the regions the XAI-driven feature selection identifies as most discriminative.

The bilateral cheek regions (right cheek + left cheek) capture the most prominent visual indicator of stroke: unilateral facial drooping. When a stroke affects the motor cortex or facial nerve pathway, it causes asymmetric weakness in the mid-face region, resulting in observable drooping on the affected side, indicating that the feature selection process is based on identifying genuine asymmetry rather than natural facial variations.

The lip regions are critical for both visual assessment and functional speech articulation. Clinically, stroke patients with facial weakness exhibit difficulty elevating one corner of the mouth when smiling, creating a characteristic asymmetric smile that is the primary visual cue in the FAST protocol. Furthermore, perioral weakness directly impacts the production of bilabial consonants (p, b, m) and labiodental consonants (f, v), contributing to dysarthric speech patterns [49].

The eyebrow and eyelid regions provide additional diagnostic value by enabling differentiation between central (upper motor neuron) and peripheral (lower motor neuron) facial palsy. In stroke-related central facial palsy, the forehead muscles are typically spared due to bilateral cortical innervation, whereas lower facial muscles show unilateral weakness. The method's selection of both upper facial landmarks (eyebrows) and lower facial landmarks (cheeks, lips, eyelids) enables comprehensive assessment of the facial nerve pathway, helping to distinguish stroke-related facial weakness.

Overall, it can be observed that the feature selection method, which aggregates importance scores from three distinct classifiers (RF, XGBoost, and CB) over five randomized train–test splits, consistently identifies clinically relevant regions. The convergence of all three classifiers on the same facial regions provides additional evidence that these landmarks capture fundamental diagnostic patterns rather than dataset-specific artifacts.

For the audio modality, the feature selection pipeline identifies the top 116 out of 128 MFCC coefficients, discarding 12 features (MFCC 125, 110, 73, 4, 122, 105, 127, 74, 23, 17, 121, and 75). This selection provides clinically interpretable insights into which spectral characteristics of speech are most affected by stroke-related dysarthria. The discarded MFCC features are primarily very high-order coefficients (e.g., MFCC 125, 127, 122, 121) that correspond to fine-grained spectral noise and components exhibiting high variability even in healthy speech. These features contribute minimal diagnostic information and may introduce noise into the classification process. Conversely, the

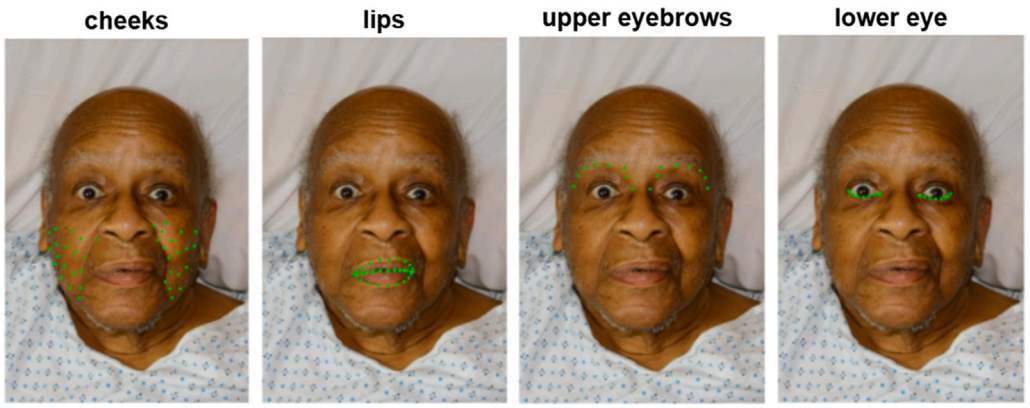


Fig. 5. Visualization of the top 10 most diagnostically relevant facial regions identified by the proposed feature selection framework. Anatomically symmetric regions are grouped together for clarity. The top regions include **cheeks** (right cheek (19 landmarks), left cheek (19 landmarks)), **lips** (upper outer lips (11 landmarks), upper inner lips (11 landmarks), lower inner lips (11 landmarks), lower outer lips (11 landmarks)), **upper eyebrows** (right upper eyebrow (eight landmarks), left upper eyebrow (nine landmarks)), **lower eye** (right lower eye (nine landmarks), left lower eye (nine landmarks)).

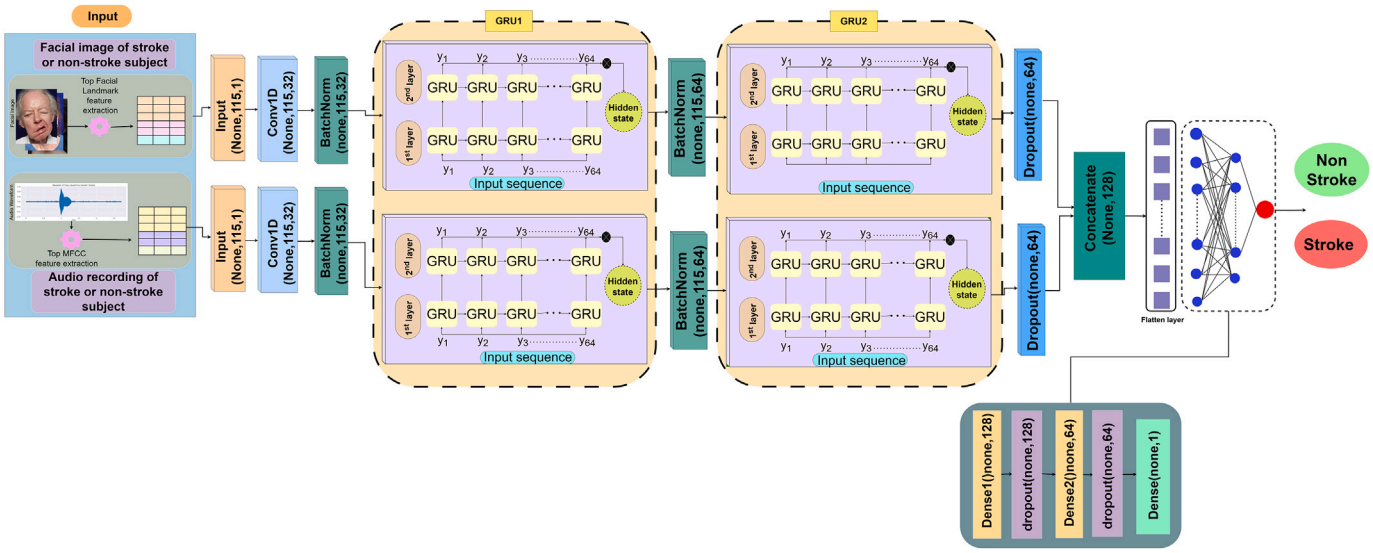


Fig. 6. Proposed multimodal hybrid architecture integrating facial landmarks and audio MFCC features with GRU-based fusion for stroke classification.

retained 116 MFCC features capture the fundamental acoustic signatures of dysarthria that are clinically observable by speech-language pathologists.

3.5. Proposed hybrid multimodal deep learning architecture

This study proposes a novel hybrid deep learning architecture designed to classify whether an individual has suffered a stroke, based on two different modalities of input: facial landmarks and audio MFCC. The proposed model combines visual and auditory information, capturing complementary features from both sources. The model has two branches dedicated to each input type (facial landmarks and MFCC features). Each branch leverages a combination of CNNs and GRUs, followed by the concatenation of both branches and the application of fully connected layers for the final classification.

The first branch processes the facial landmarks, consisting of layers to extract spatial and temporal patterns from the data. Initially, the input sequence passes through a 1D convolutional layer with 32 filters and a kernel size 3. Next, a batch normalization layer is applied for stabilization. The output is then passed through a GRU layer to capture temporal dependencies in the sequence of facial landmark points, with the first GRU layer having 64 units. Afterward, another batch normalization layer is applied, followed by the second GRU layer, which consists of 64 units. Finally, a dropout layer with a rate of 0.3 is added to mitigate overfitting by randomly deactivating a fraction of neurons during training. The second branch processes the MFCC features similarly, using a 1D convolutional layer, batch normalization, two GRU layers, and a final dropout layer with a rate of 0.3.

The outputs of both branches are combined using a concatenation layer. Subsequently, the merged features are passed through two fully connected dense layers. The first dense layer consists of 128 neurons with ReLU activation, followed by a dropout layer with a rate of 0.4. The second dense layer, containing 64 neurons and also using ReLU activation, is followed by another dropout layer with a rate of 0.4. Finally, the output layer has a single neuron with a sigmoid activation function, providing a binary prediction. Fig. 6 shows a detailed illustration of the proposed hybrid multimodal architecture.

3.6. Hyperparameter tuning

Table 2 shows the top-ranked configurations of the CNN-GRU model, selected from an initial set of 128 configurations based on empirical analysis. The results reveal that consistently using 32 convolutional filters and a learning rate of 0.001 yields high accuracy across different setups. Models with larger GRU layers (64 units) and a two-stage dense layer paired with moderate dropout rates (0.3–0.4) achieve

Table 2
Top Ranked Configurations of the CNN-GRU Model Based on Accuracy.

Rank	Accuracy	Configuration
1	0.9533	Conv=32, GRU1=64, GRU2=64, Dense1=128, Dense2=32, Dropout=0.3, LR=0.001
2	0.9469	Conv=32, GRU1=32, GRU2=32, Dense1=128, Dense2=64, Dropout=0.4, LR=0.001
3	0.9462	Conv=32, GRU1=32, GRU2=32, Dense1=128, Dense2=32, Dropout=0.4, LR=0.001
4	0.9462	Conv=32, GRU1=32, GRU2=32, Dense1=64, Dense2=64, Dropout=0.3, LR=0.001
5	0.9440	Conv=32, GRU1=64, GRU2=32, Dense1=128, Dense2=32, Dropout=0.4, LR=0.001
6	0.9411	Conv=32, GRU1=32, GRU2=64, Dense1=128, Dense2=64, Dropout=0.4, LR=0.001
7	0.9390	Conv=32, GRU1=64, GRU2=32, Dense1=128, Dense2=32, Dropout=0.3, LR=0.001
8	0.9256	Conv=32, GRU1=32, GRU2=32, Dense1=64, Dense2=32, Dropout=0.4, LR=0.001
9	0.9233	Conv=32, GRU1=32, GRU2=64, Dense1=64, Dense2=64, Dropout=0.3, LR=0.001
10	0.9226	Conv=32, GRU1=32, GRU2=64, Dense1=128, Dense2=32, Dropout=0.3, LR=0.001

the best performance, with the highest accuracy reaching 95.33%. This suggests that deeper recurrent layers combined with adequate regularization balance learning capacity and overfitting prevention effectively. Slight variations in dense layer sizes and dropout rates cause only minor accuracy drops, showing the model's robustness to hyperparameter tuning. Overall, configurations with stronger GRU layers and sufficiently dense units consistently outperform those with smaller or simpler setups, emphasizing the importance of well-designed recurrent and fully connected layers in optimizing CNN-GRU performance.

The superior performance of specific configurations can be attributed to several factors. The 32 convolutional filters effectively capture local spatial patterns in facial landmarks and MFCC sequences without introducing excessive parameters that could lead to overfitting given the dataset size of 1990 samples. Larger GRU units (64) significantly outperform smaller configurations (32) because temporal dependencies in both facial movements and speech sequences require sufficient memory capacity to model long-range sequential patterns characteristic of stroke symptoms. The two-stage dense architecture (128 to 64 neurons) with moderate dropout (0.3–0.4) achieves an optimal balance between model expressiveness and regularization, preventing overfitting while maintaining adequate capacity to learn complex multimodal feature interactions. The learning rate of 0.001 consistently demonstrates stable convergence across all tested configurations. These findings indicate that the optimal architecture balances representational capacity with appropriate regularization to effectively learn from limited multimodal data while maintaining strong generalization capability.

3.7. Epoch-wise stochastic modality masking framework

This section provides a detailed description of the proposed epoch-wise stochastic modality masking framework. The framework is explicitly designed to allow multimodal learning when facial and auditory modalities come from different sources and lack subject-level correspondence. The framework consists of two key stages: aligned data preparation and dynamic masking during model training. Unlike traditional multimodal pipelines that assume synchronized data from the same subject, the proposed approach operates under a cross-source setting, where facial landmarks and audio MFCC features are collected separately but share common diagnostic labels. To begin with, the facial landmarks and audio MFCC datasets are aligned row-wise such that both modalities share identical target labels at every index. While this does not imply sample-level correspondence, it serves as a weak form of multimodal pairing that partially simulates a multimodal data setup at the label level. Subsequently, a dynamic masking strategy is implemented that transforms both the facial landmarks and audio MFCC datasets by zero-masking a selected subset of samples within each modality, effectively imitating missing modality information. The proportion of samples that are zero-masked is varied at every epoch, resulting in a unique reconstructed dataset each time. Consequently, the model is compelled to learn unimodal, as well as multimodal representation, despite the absence of true multimodal correspondence.

3.7.1. Training and testing data preparation

The model training process begins with two distinct datasets: one consisting of facial landmark features and the other composed of audio-based MFCC features. Both datasets contain only numerical values, making them ideal for direct model input. The entire process of training data preparation is illustrated in Fig. 7. Initially, the facial landmark dataset includes 998 samples of stroke and 998 labeled as non-stroke. The MFCC dataset has 1014 non-stroke and 982 stroke instances. To avoid potential biases in the training process, each dataset is balanced by randomly undersampling the majority class, resulting in 995 stroke and 995 non-stroke instances across the facial and MFCC datasets (step 1). After balancing, min–max normalization is applied to each dataset to ??the feature values between 0 and 1 (step 2). The same steps (1 and 2) are also performed for the audio MFCC dataset. Following this, a label-wise alignment of the facial landmarks and audio MFCC for stroke and non-stroke cases is performed (step 3). This ensures that corresponding rows in both datasets represent the same class instance, thus ensuring diagnostic consistency across modalities. For instance, if the sample at row 3 in the facial landmark dataset represents a stroke case, then the sample at the same row in the MFCC dataset also corresponds to a stroke case. Likewise, if row 100 in the landmark data refers to a non-stroke individual, the MFCC dataset at that same position also reflects a non-stroke instance. This ensures that the target variables (L_Stroke and M_Stroke) or ground truth columns are exactly identical. Here, shared labels act as a weak supervisory signal enabling cross-modal learning under heterogeneous data conditions.

Next, the feature sets of each dataset are separated from the target variables, and a single common target variable column is considered for both modalities of input. As both datasets are already aligned in terms of target labels, only one of the target columns is sufficient for the split. Then, the facial landmark features, the audio MFCC features, and the target variable vector are split into training (70%) and testing (30%) sets (step 4), while strictly maintaining label alignment across both modalities. Moreover, for the dysarthria dataset, a speaker-independent split is applied to prevent speaker leakage. Table 3 shows the division of speakers and sample counts for training and testing of the audio dataset.

Finally, a total of six separate files are obtained: the facial landmark training set, the MFCC training set, the facial landmark test set, the MFCC test set, and two corresponding target label sets, one for the training data and one for the test data, shared across both modalities.

3.7.2. Model training loop with dynamic modality masking

The designed model can handle three different types of input data: only facial landmarks, only MFCC features, or a combination of both provided simultaneously. To enable the model to learn and handle varying modalities, a data mixing function is implemented, which creates a new batch of training data for each epoch by combining samples from three different sets: full modality (both facial and MFCC features), landmarks-only, and audio MFCC-only inputs. Fig. 8 illustrates the stepwise workflow for model training.

Table 3
Speaker-wise division of audio samples for training and testing in the audio MFCC dataset.

Class	Test speakers	Test samples	Training speakers	Training samples
Female Non-Dysarthria	FC01	164	FC03	350
Female Dysarthria	F01	132	F03	350
Male Dysarthria	M03	150	M04 + M05	350
Male Non-Dysarthria	MC03	150	MC01 + MC02 + MC04	350

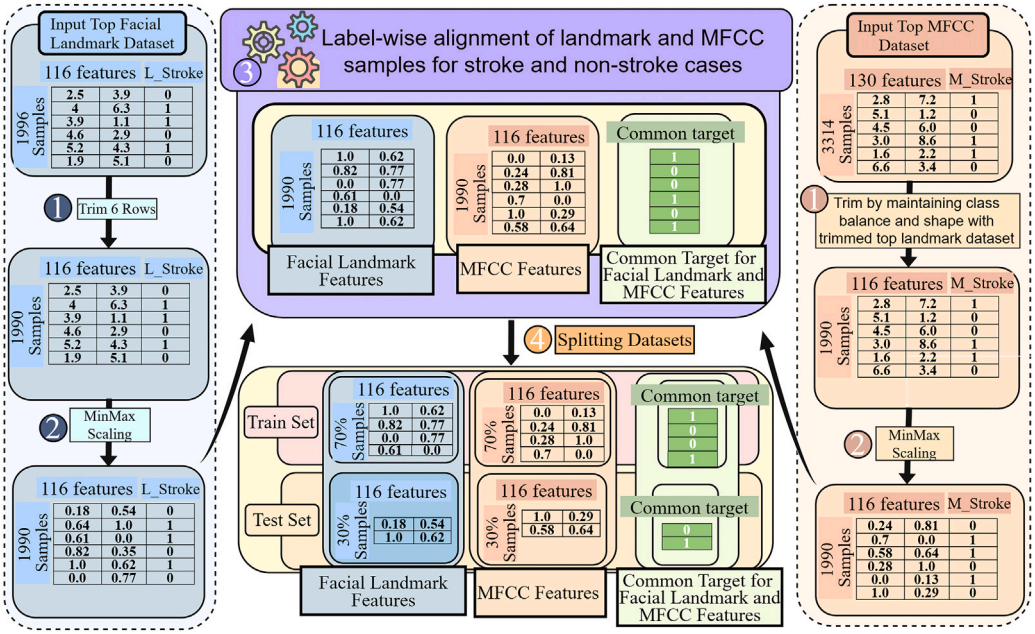


Fig. 7. Schematic of the training and testing data preparation pipeline for multimodal stroke classification.

During each epoch, the data mixing function (*generate_mixed_data*) takes the landmarks train set and audio MFCC train set as inputs. Subsequently, it generates a dynamic ratio, which is then applied to both inputs. Each dataset is divided into three subsets, ensuring a similar split and alignment of the corresponding subsets of landmarks and audio MFCC. Although the ratio changes with every epoch, the summation is always one. For example, in a particular epoch, the data might be split following a 4:3:3 ratio and a 6:3:1 ratio in another. The ratios are created on the basis of Eq. (11).

$$a : b : c \quad \text{such that} \quad a + b + c = 1 \tag{11}$$

For a given epoch, as mentioned earlier, the facial landmarks train set is divided into three separate subsets based on Eq. (11), such that the first subset contains proportion a of the total data, the second subset comprises proportion b, and the third consists of proportion c. Moreover, all feature values within the third subset (corresponding to proportion c) are transformed to zero, making those instances invalid facial landmark features.

Similarly, the audio MFCC train set is divided into three subsets based on the same ratio ($a : b : c$). In contrast to the landmark train set, where all features in the third subset (proportion c) are assigned zero values, the feature values in the second subset (proportion b) of the audio MFCC train set are transformed to zero.

In the following step, the three subsets of the landmark train set, initially partitioned in the ratio $a : b : c$, are vertically concatenated to form a new dataset named the landmark mixed train set, which has the same structure as the original dataset. However, in the reconstructed dataset, the segment corresponding to the c-proportion consists entirely of zero values. Likewise, the subsets of the audio MFCC train set are vertically concatenated to create the audio MFCC mixed train set, but in this case, the subset corresponding to proportion b contains only zero values. Fig. 9 provides an overview of the general approach from partitioning to transformation and reconstruction. It illustrates the initial state of the training data, followed by the partitioning process based on a specific ratio ($a : b : c$), the transformation of the feature values of a particular partition, and the final state of the dataset after reconstruction through vertical concatenation.

The newly formed landmark mixed train set and audio MFCC mixed train set are then passed as input into the first and second branches of the proposed model, respectively, at a given epoch. Based on the composition of these two proportioned datasets, the model encounters a varying combination of modalities during training. Specifically, the following cases are encountered:

- **Case 1**: Both the landmark mixed train set and the audio MFCC mixed train set instances contain valid facial features and acoustic features.
- **Case 2**: The landmark mixed train set instance contains valid facial features, but the audio MFCC mixed train set instance contains only zero values.
- **Case 3**: The landmark mixed train set instance contains only zero values, while the audio MFCC mixed train set instance contains valid acoustic features.

Cases 1, 2 and 3 are illustrated in Fig. 10. The Figure represents the landmark mixed train set and the audio MFCC mixed train set, along with their subdivisions based on the proportions. It also shows how these proportions vary across different epochs, highlighting the dynamic ratio based approach used in this study.

For epoch 1, the subdivision labeled as both landmark & audio MFCC input (consisting of the full landmark set and the full audio MFCC set), as shown in Fig. 10, represents Case 1. In this scenario, the model receives both facial and acoustic features as input simultaneously, enabling it to learn how to make predictions based on a complete

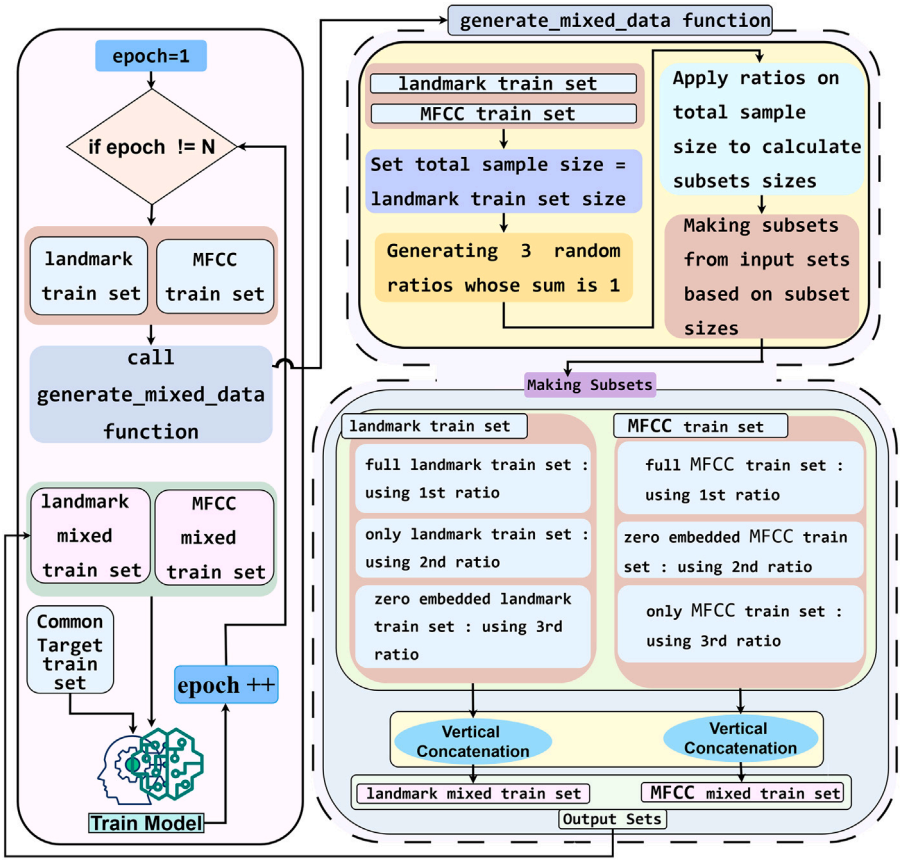


Fig. 8. Workflow overview of the model training loop with dynamic modality mixing across epochs.

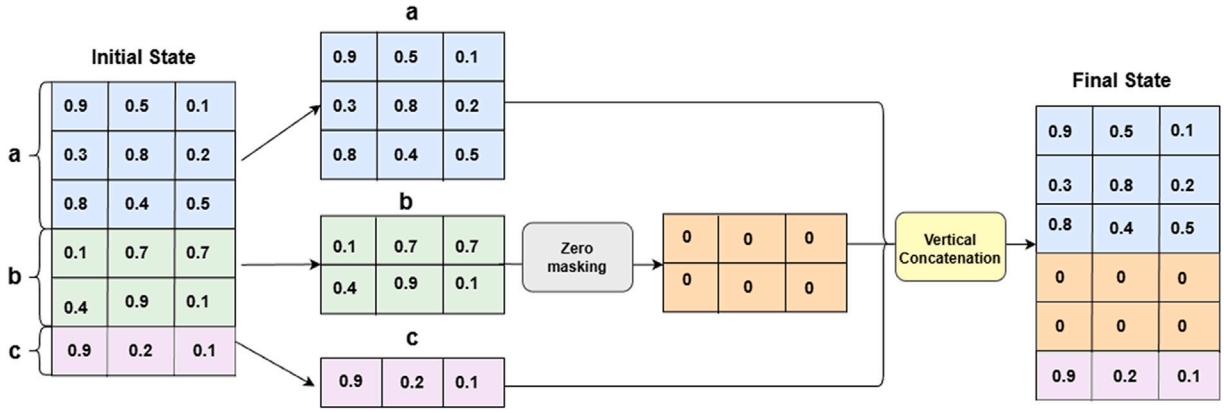


Fig. 9. Partitioning $(a : b : c)$, zero-masking of a certain partition (b), and reconstruction of the training data.

set of multimodal data The subdivision labeled landmark only input, consisting of the only landmark set combined with the zero-embedded audio MFCC set, corresponds to Case 2. Here, the model receives valid facial features but no acoustic information (audio MFCC values are all zero), effectively training the model to diagnose stroke based only on facial features. Case 3 is represented by the subdivision labeled as audio MFCC only input, comprising of zero embedded landmark set and only audio MFCC input (as illustrated in Fig. 10). This configuration removes facial features from the input data by replacing the feature values with zeros; therefore, training the model to make predictions solely on the acoustic (audio MFCC) features.

While the ratio is generated randomly during each epoch, empirical analysis revealed that the landmark only scenario produced the weakest model performance during testing. Therefore, the stochastic ratio generation is designed to slightly bias the masking process toward creating a larger subset of landmark only instances, ensuring the model is exposed to this case more often during training.

The training performance plots, as shown in Fig. 11, depict a strong and consistent learning trajectory over 600 epochs. The plot to the left, Accuracy Over Epochs, displays an accuracy value that initially stands at close to 55% but moves steadily upwards to reach about 95%. That shows the model gets more data over time, therefore it improves classification. The upward trend starts to slow down after epoch 400.

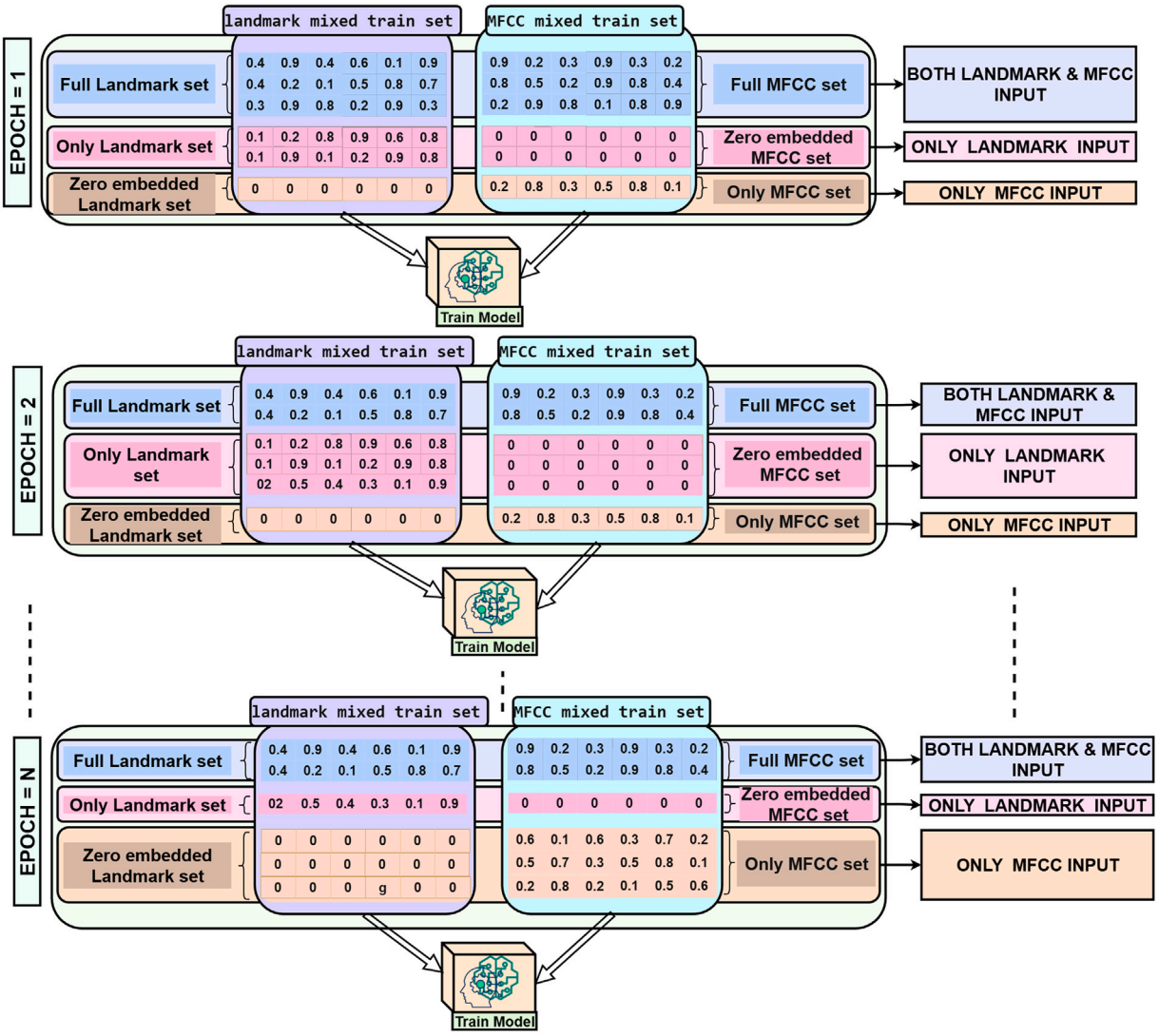


Fig. 10. Illustration of varying modality inputs (facial landmarks and audio MFCC) during training across epochs.

This reflects the fact that the model is close to convergence, and further improvements are small. On the plot to the right, Loss Over Epochs, the loss starts from around 78% and drops sharply at the beginning phase showing the modeling phase of the fast pattern capture of the data. After epoch 100, the loss further continues downward with few upward fluctuations indicating probably drops in learning rate or the variance of some small-sized batches but still trending downwards till its value reaches below 10%. An inverse relationship between accuracy and loss is very smooth, thus affirming that the training is stable. Importantly, the curves remain smooth without abrupt fluctuations, indicating stable learning and no signs of overfitting.

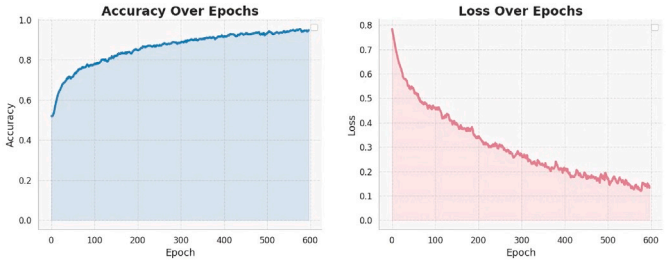


Fig. 11. Training Accuracy and Training loss over epochs.

3.8. Model evaluation

This section provides a thorough description of the overall testing procedure, along with the evaluation metrics used to assess performance of the model.

3.8.1. Model testing strategy

To evaluate the proposed novel multimodal model, a rigorous systematic scheme is designed with comprehensive testing strategy encompassing three distinct test cases shown in Fig. 12. The test cases are as follows:

Test case 1 (Full multimodal input): In this scenario, the model is provided with both the landmark test set and the MFCC test set. This case serves as a baseline to evaluate the model's performance when complete multimodal information is available. It reflects the ideal condition for the model and is expected to yield the highest accuracy.

Test case 2 (Landmark Only (audio MFCC Embedded with Zero)): Here, the landmark test set is provided, while the audio MFCC input is replaced with zero embeddings to simulate the absence of audio features. This case is designed to test the model's ability to rely solely on visual (landmark) features for inference. It helps

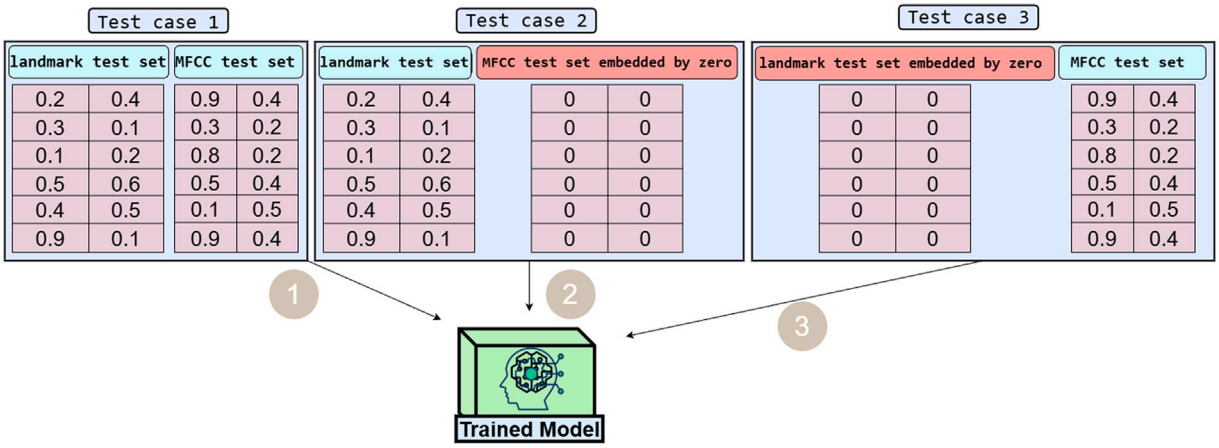
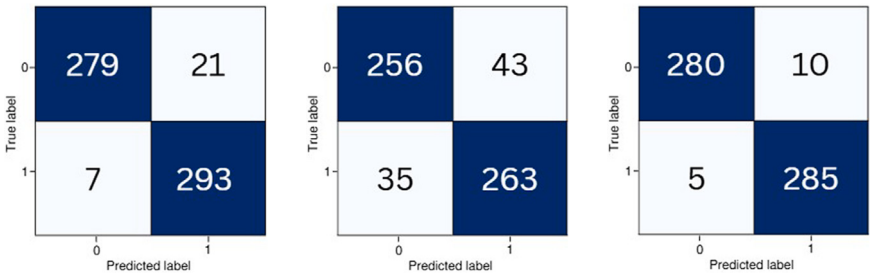


Fig. 12. Illustration of varying modalities of input during testing.



(a) Both facial landmarks and MFCC (1: Stroke; 2: Non-Stroke).

(b) Only facial landmarks (1: Stroke; 2: Non-Stroke).

(c) Only audio MFCC (1: Stroke; 2: Non-Stroke).

Fig. 13. Confusion matrices of the proposed model across all three data types.

quantify the standalone contribution of landmark data to the model's decision-making.

Test case 3 (audio MFCC Only (Landmark Embedded with Zero)): In this configuration, the audio MFCC test set is used alongside a zero-embedded landmark input, thereby simulating the absence of facial landmark data. This test case evaluates how well the model performs with only audio features and complements the insights gained from Test Case 2.

This testing strategy provides a thorough assessment of the model under both ideal and constrained conditions. It provides insights into the model's ability to make accurate predictions when only one modality is present and validates the effectiveness of the dynamic multimodal training approach in real-world scenarios.

3.8.2. Performance metrics

The machine learning algorithms used in this research are evaluated using metrics such as accuracy, precision, recall, and F1-score.

Precision: Positive Predictive Value, also known as precision, as shown in Eq. (12), refers to the proportion of true positive (TP) predictions out of all instances that are predicted as positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{12}$$

Recall: Recall, also known as sensitivity, is defined as the proportion of actual positive cases that are correctly identified. A higher recall rate indicates that more positive instances are successfully detected. The formula for calculating recall is shown in Eq. (13).

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{13}$$

F1-score: F1-score is regarded as a more dependable performance metric because it combines both precision and recall into a single value by calculating their harmonic mean. It provides a balanced assessment of a model's accuracy, especially when dealing with imbalanced data. The value of F1-score ranges from 0 to 1 and is defined in Eq. (14).

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{14}$$

Accuracy: Accuracy is a commonly used metric that evaluates overall model performance by calculating the proportion of total predictions that are correctly classified. Eq. (15) shows the formula for calculating accuracy in terms of TP, true negatives (TN), false positives (FP), and false negatives (FN).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{15}$$

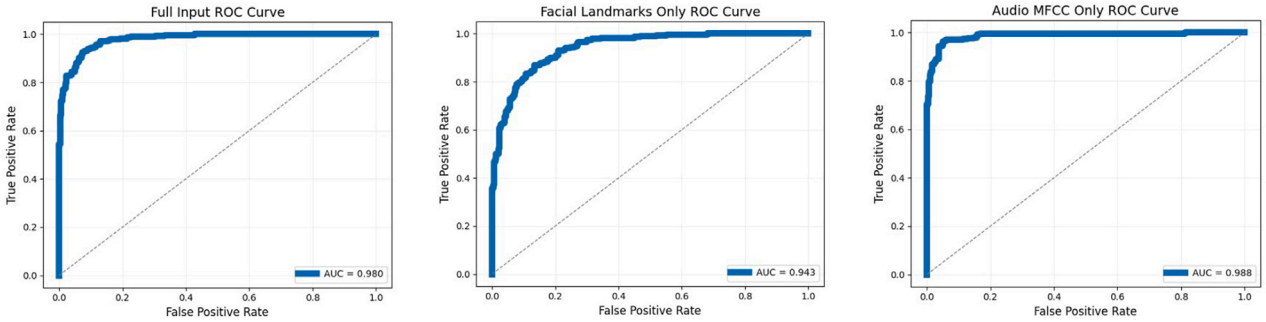
4. Results analysis

Table 4 summarizes the performance of the evaluated model architectures across three testing scenarios: combined facial landmarks and audio MFCCs, facial landmarks only, and audio MFCCs only. The evaluated configurations include standalone models such as CNN and LSTM, as well as hybrid architectures including CNN-LSTM, CNN-Transformer, CNN-BiLSTM, DNN-LSTM, CNN-LSTM & CNN-Transformer, and CNN-GRU.

In the multimodal setting (combined facial landmarks and audio MFCCs), the CNN-GRU achieves an accuracy of 95.33%, precision of 93.31%, recall of 97.67%, and an F1-score of 95.44%. Closely following are CNN-BiLSTM (accuracy: 94.78%) and CNN-LSTM (accuracy: 94.69%), indicating that hybrid CNN–recurrent architectures are particularly effective at learning complementary spatial and temporal representations from multimodal data. The standalone CNN model also performs competitively with an accuracy of 95.84%, while purely recurrent or fully connected hybrids such as LSTM (92.16%) and DNN-LSTM

Table 4
Model Performance Across Modalities (Face and Audio, Face Only, Audio Only).

Model	Combined facial landmarks and audio MFCC test				Facial landmarks only test				Audio only test			
	Accuracy	Precision	Recall	F1 Score	Accuracy	Precision	Recall	F1 Score	Accuracy	Precision	Recall	F1 Score
CNN	0.9584	0.9627	0.9619	0.9623	0.5342	0.6721	0.6413	0.6463	0.9658	0.9665	0.9659	0.9662
LSTM	0.9216	0.9442	0.9431	0.9436	0.6918	0.7510	0.7472	0.7461	0.9284	0.9396	0.9388	0.9392
CNN-LSTM	0.9469	0.9511	0.9504	0.9507	0.7404	0.7598	0.7588	0.7586	0.9582	0.9589	0.9581	0.9585
CNN-TRANSFORMER	0.9342	0.9409	0.9401	0.9405	0.5008	0.2496	0.5000	0.3334	0.9257	0.9264	0.9256	0.9260
CNN-BiLSTM	0.9478	0.9526	0.9519	0.9522	0.8442	0.7820	0.7806	0.7803	0.9672	0.9679	0.9671	0.9675
DNN-LSTM	0.9441	0.9496	0.9489	0.9492	0.7152	0.7262	0.7104	0.7052	0.9564	0.9571	0.9563	0.9567
CNN-LSTM & CNN-TRANSFORMER	0.9393	0.9431	0.9424	0.9427	0.7638	0.7571	0.7571	0.7571	0.9431	0.9440	0.9432	0.9436
CNN-GRU	**0.9533**	**0.9331**	**0.9767**	**0.9544**	**0.8687**	**0.8595**	**0.8826**	**0.8709**	**0.9741**	**0.9661**	**0.9828**	**0.9744**



(a) Both facial landmarks and MFCC. (b) Only facial landmarks. (c) Only audio MFCC.

Fig. 14. ROC curves of the proposed model across different input modalities.

(94.41%) show comparatively weaker performance. Transformer-based architectures, including CNN-TRANSFORMER (93.42%) and the hybrid CNN-LSTM & CNN-TRANSFORMER (93.93%), exhibit a modest reduction in accuracy, suggesting limited benefit from self-attention mechanisms in this multimodal context.

A pronounced performance degradation is observed when models are evaluated using only facial landmarks. In this unimodal setting, CNN-GRU again demonstrates the strongest performance, achieving an accuracy of 86.87% and an F1-score of 87.09%, followed closely by CNN-BiLSTM (accuracy: 84.42%). Most other architectures experience substantial declines, most notably the standalone CNN (53.42%) and CNN-TRANSFORMER (50.08%), which perform only marginally better than random guessing. These results indicate that facial landmarks alone provide insufficient discriminative information for many architectures, particularly those without robust temporal modeling.

Conversely, in the audio MFCC–only evaluation, all models achieve consistently high performance, highlighting the strong discriminative power of audio features. The CNN-GRU again attains the highest accuracy at 97.41%, followed by CNN-BiLSTM (96.72%) and CNN (96.58%). Even transformer-based and fully connected hybrid models maintain strong results, with accuracies exceeding 92%, and only minor performance differences observed across architectures. This consistency suggests that most models are well-suited to extracting salient temporal patterns from audio representations.

The proposed hybrid CNN-GRU-based model demonstrated the most balanced performance, benefiting from convolutional layers for local feature extraction and recurrent layers to model the temporal structure of the data. In contrast, transformer-based variants performed inconsistently and suffered significant drops in accuracy when one modality is absent, highlighting their sensitivity to missing data. The results provide evidence that multimodal CNN-GRU achieves high accuracy.

Fig. 13 highlights the confusion matrices of the proposed model across all three data variations, including both facial landmarks and audio MFCC 13(a), only facial landmarks 13(b) and only audio MFCC 13(c). In the case of both facial landmarks and audio MFCC, the model shows the most optimal results, with 279 TNs and 293 TPs, while making only 21 FPs and 7 FNs, suggesting that the combination of both modalities leads to reliable predictive performance. In contrast, when only facial landmarks are provided, performance drops noticeably, with a total of 43 FPs and 35 FNs. However, using only audio MFCC as input produces better results, almost as strong as the combined case, with 280 TNs, 285 TPs, 10 FPs and 5 FNs. The findings suggest that audio features are highly effective at capturing patterns relating to stroke on their own, matching the performance of the multimodal approach, whereas relying only on visual indicators substantially increases the risk of potential misdiagnosis.

The ROC curves in Fig. 14 depicts the proposed model under three input modalities. The model achieves an AUC of 0.98 when both modalities are provided 14(a), indicating extremely high classification capability. On the other hand, when only facial landmarks are given, the curve becomes flatter, yielding a lower AUC of 0.943 (Fig. 14(b)), corresponding to a higher risk of misclassifications. Alternatively, the use of MFCC results in a curve that is almost identical to the multimodal case (Fig. 14(c)) with AUC of 0.988, further confirming the reliability of audio features for stroke detection.

4.2. Structural analysis of data with HDBSCAN

The HDBSCAN clustering visualizations in Figs. 15 and 16 show how the latent feature representations from multimodal (MFCC and facial landmarks) input features are organized into separable groups.

In Fig. 15, the visualization represents features based on facial landmarks, which demonstrate two stark clusters in the 2D projection and the 3D PCA space, reflecting separability between yellow and green clusters. This indicates that the geometric cues encoded in facial landmarks offer a structured data representation. However, despite being compact, the facial embeddings is limited, resulting in lower classification accuracy compared to audio-based features. Yet, the clustering still helpfully organizes the data, with a few outliers on the edges, which represent errant facial configurations or noise artifacts due to the web-based data collection.

Fig. 16 portrays the clusters of audio MFCC features; although it presents two clear clusters, they are not as distinctly segmented as the facial embeddings (Fig. 15). The clusters appear less compact and more dispersed, with overlapping boundaries and elongated groupings. Despite this, the MFCC features achieve higher accuracy, suggesting

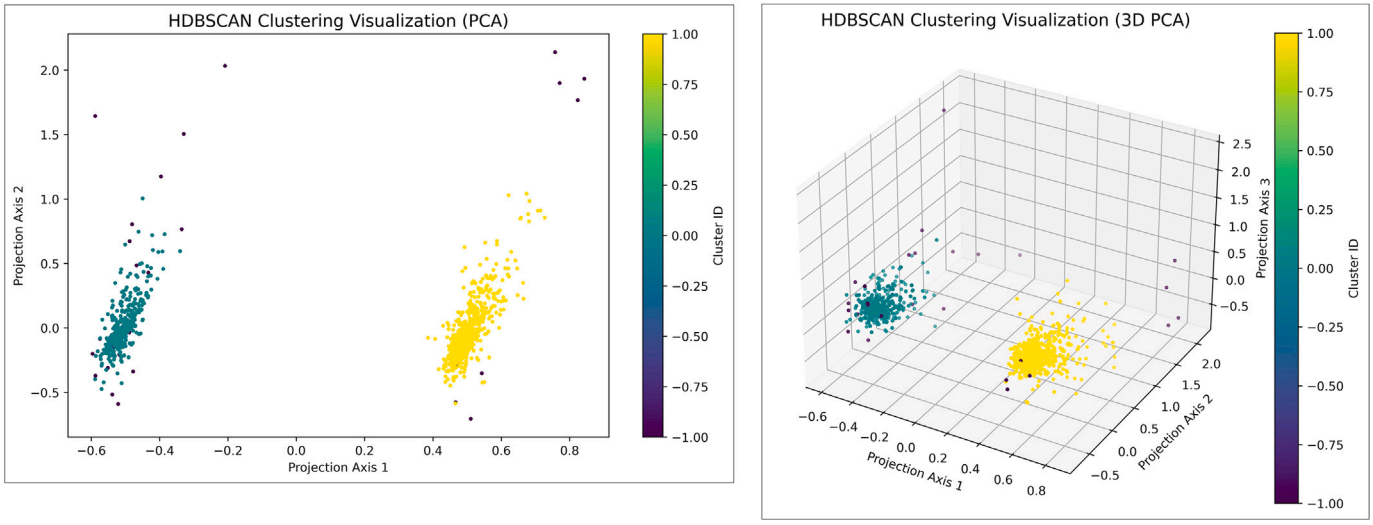


Fig. 15. 2D and 3D visualization of HDBSCAN clustering on facial features.

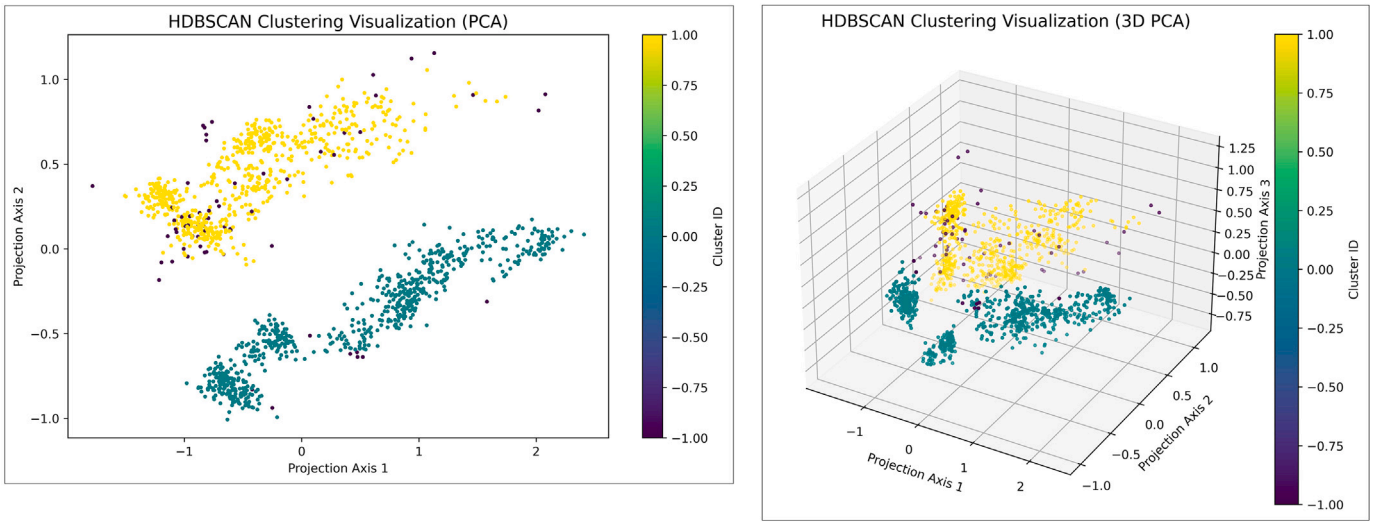


Fig. 16. 2D and 3D visualization of HDBSCAN clustering on MFCC features.

that the acoustic cues carry richer information, capturing essential characteristics such as pitch, tone, and prosody, which are highly effective for classification tasks.

Overall, the HDBSCAN analysis confirms that compact clustering does not necessarily translate into higher discriminative performance, as evident in facial embeddings. On the contrary, while MFCC features are noisier in clustering, they carry stronger classification signals. These findings further support the rationale for adopting a multimodal approach, where each modality compensates for the limitations of the other, serving as a robust diagnostic framework.

4.3. Comparison with state of the art approaches

Table 5 shows a comparative analysis of the proposed CNN-GRU model against several other state of the art methods across three modalities, including facial images, audio, and the combination of facial images and audio.

For the facial image only, the highest accuracy is reported by [22], achieving 96.8%, closely followed by [27] at 95% and [16] at 94.29%. In comparison, the proposed CNN-GRU attains 86.87%, which is slightly lower than the best-performing approaches. In the audio-only setting, the proposed model achieves an accuracy of 97.41%, demonstrating performance that is either superior to or comparable with existing state of the art approaches, including 93.97% reported in [32] and 98% reported in [31].

A similar trend is observed in the multimodal setting. The proposed CNN-GRU model achieves an accuracy of 95.33%, surpassing the methods reported in [36,39], and [40], where the accuracies are 95.0%, 82.6% and 80.85%, respectively.

Overall, the results suggest that although the proposed model slightly under-performs in the facial image only modality, it outperforms all other state of the art techniques in both audio and multimodal scenarios, highlighting its strength in leveraging complementary cross-modal features.

4.4. Model robustness

This section provides a detailed analysis of model performance under corrupt and degraded input conditions, simulating realistic scenarios that the model is likely to encounter in practical deployment. Initially, the unimodal capabilities of the model are tested by providing it with corrupted facial images or audio recordings, while the other modality is masked. Following this, the multimodal configuration is

Table 5
Performance Analysis (accuracy) with Existing State-of-the-art Models.

Reference	Model	Facial images	Audio	Facial images and audio
[16]	Comparison of CNN, RNN, LSTM, GRU	94.29%	–	–
[22]	Kinect V2 sensor to track Facial Animation Units (FAUs) dynamic markers to encode facial muscle movements	96.8%	–	–
[27]	Facial landmarks combined with decision tree classifiers	95%	–	–
[31]	Hybrid CNN-SVM model	–	98%	–
[32]	Classifier using MFCC, CQCC, speech-specific, and i-vector features	–	93.97%	–
[36]	ResNet50 (video) and VGGish (audio) fused using an Asymmetrical Multi-Modal Attention (AMMA) mechanism	–	–	82.6%
[40]	"M3Stroke", multimodal encoder (M3Encoder) for video frames and audio spectrograms, ResNet (video), Audio Spectrogram Transformer (AST) (audio)	–	–	80.85%
[39]	"StrokeSave", a combination of CNN-RNN, AAM, and SVM	–	–	95%
Proposed Model	**CNN-GRU**	**86.87%**	**97.41%**	**95.33%**

Table 6
Facial image channel performance under different corrupted conditions.

Corruption type	Accuracy (%)
Brightness & Contrast Variation	87.40
Gaussian Noise	87.12
JPEG Compression	87.00
Shadow Occlusion	85.66
Motion Blur	86.34
Mean Accuracy (Individual Corruptions)	86.30
All Corruptions (Mixed)	86.94

tested by simultaneously distorting both facial and audio inputs, identifying the robustness of the system when both input modalities are affected.

4.4.1. Facial input channel robustness evaluation

To simulate visual variations in real world imaging conditions, the facial images are subjected to common imaging distortions such as changes in lighting (brightness and contrast), sensor or transmission artifacts (Gaussian noise), loss of quality (JPEG compression), partial feature masking due to shadows (shadow masking) and effect of subject or camera movement (motion blur). For testing, a total of 600 facial images are selected. The model is evaluated on each corruption scenario individually, with all 600 samples subjected to the same type of distortion. Additionally, a mixed corruption test is conducted, where 120 samples from each corruption type is combined to form a final test set.

Table 6 presents the model's performance for each individual corruption type, as well as for the mixed corruption scenario. From the data, it can be observed that the model maintains stable performance across different visual distortions, with accuracies ranging from 85.66% to 87.40% for the individual corruptions. The highest accuracy (87.40%) is achieved under brightness and contrast variations, suggesting the model is relatively resilient to changes in lighting conditions. Performance slightly lowers (85.66%) for shadow occlusion, indicating that partial masking of facial features has a notable impact. This is expected, as the model may miss relevant facial landmarks important for stroke detection. Other corruptions such as Gaussian noise, JPEG compression, and motion blur yield intermediate accuracies (87.12%, 87.00%, and 86.34%, respectively), highlighting the model's robustness against common sensor and motion-related distortions. When tested against the mixed corruption test set, the model achieves an accuracy of 86.94%, slightly higher than the mean performance (86.30%) of the model across all individual corruptions. This shows the model can effectively maintain reliable performance across various practical conditions, reflecting its suitability for deployment in outdoor or uncontrolled environments, where imaging conditions are often dynamic.

4.4.2. Audio channel robustness evaluation

The audio data is subjected to different real-world acoustic distortions such as reverberation (echoes in enclosed or large spaces), low pass and high pass filtering (variations in microphone response and signal transmission, babble noise (interference from multiple speaker) and additive white noise (random environment or sensor noise). Similar to the facial data, a test set of 600 samples is created. Additionally, to prevent any speaker leakage, the test set is constructed to ensure that no speakers overlap with those in the training set. Likewise, the model is assessed on individual corruption scenarios, as well as on a mixed test set comprising 120 samples from each type of distortion.

From Table 7, it is observed that the model maintains a stable performance across different acoustic distortions, with accuracy scores between 89.95% and 97.65%. The best score is observed for babble noise (97.65%), indicating the model maintains performance even in the presence of overlapping speech. Conversely, the lowest accuracy occurs for low-pass filtering (89.95%), suggesting that the loss of information due to filtering high frequencies has an adverse impact on performance. However, in the case of high-pass filtering, where low frequencies are reduced, the accuracy increases to 93.66%. This suggests that the low-frequency components of speech are particularly important for detecting stroke-related patterns. Reverberation and high-pass result in even higher scores (95.33% and 93.66%, respectively). In the case of

Table 7
Audio channel performance under different corrupted conditions.

Corruption type	Accuracy (%)
Reverberation	95.33
Low-pass Filtering	89.95
High-pass Filtering	93.66
Babble Noise	97.65
Additive White Noise (SNR-controlled)	96.24
Mean Accuracy (Individual Corruptions)	94.57
All Corruptions (Mixed)	96.47

the mixed corruption test set, the model achieves 96.47% accuracy, surpassing the mean accuracy across individual corruptions. Overall, the results suggest that the model is effective across diverse and dynamic acoustic environments.

4.4.3. Multimodal robustness evaluation

Finally, to evaluate the model's performance when both modalities are simultaneously corrupted, the mixed corrupted test sets of facial and audio data are provided to their respective branches. In this scenario, the model achieves an accuracy of 94.97%, demonstrating its effectiveness in dynamic multimodal environments.

4.5. Interpretation of overall results

From the analysis, it is evident that audio-only models far outperform face-only models, with the CNN–GRU model achieving 97.41% accuracy in the audio-only settings compared to 86.87% with facial data alone). This performance gap exists largely due to the influence of dysarthric speech on spectral and prosodic cues that are captured in MFCC embeddings and are useful to provide clear separability in latent space versus facial asymmetry. In contrast, facial asymmetry is often subtle and can be affected by motion,lighting, and demographic factors. While the audio-only achieves near-perfect accuracy, perfect speech is not always available in real world cases. Distortions such as aphasia, intubation, or environmental noise can affect the speech quality, causing a unimodal audio model to fail catastrophically. The multimodal CNN–GRU, however, benefits from complementary features from both modalities: speech provides a strong discriminative perspective, while face landmarks can compensate when audio is missing or corrupted. Therefore, this fusion strengthens two things, higher peak accuracy (95.33% overall), and a degree of resilience to real-world data variations. The use of stochastic modality masking further enhances the model's resilience by allowing it to learn from both unimodal and multimodal inputs, thus allowing it to degrade gracefully when one modality is missing. The benefit of this approach is particularly evident when comparing different architectures: the CNN–GRU maintains strong performance even with a single input stream, whereas transformer-based models show a substantial drop in accuracy, indicating susceptibility to missing data This observation suggests that temporal gating architectures trained with modality masking are especially suitable for pre-hospital or low-resource settings where data may be incomplete or unreliable. Moreover, linearly separable samples consistently form boundary regions in the HDBSCAN plots (Figs. 15 and 16), indicating that the remaining errors are primarily due to clinically challenging cases where even human raters may disagree. Comparisons with existing strategies (Table 5) show that the CNN–GRU achieves accuracies comparable to or exceeding those of previous multimodal stroke detection systems.

Finally, when the proposed training strategy is tested with corrupted or incomplete inputs, the model continues to perform reliably, highlighting the robustness of the framework. This suggests that the approach is not only highly accurate but also well-suited for real-world deployment

In summary, unimodal models only succeed when the target symptom is obviously present and quantifiable (i.e., strong dysarthria or clear facial droop), and they often fail when subtlety, noise, or data are missing. But multimodal fusion succeeds (especially CNN–GRU and stochastic masking), because it can take advantage of complementary cues and tolerate missing modalities, and can allow for residual errors to come from clinically ambiguous cases as opposed to random misclassification. The triangulation of accuracy, robustness, and interpretability represent a step closer to stroke detection systems that are practical and deployable.

5. Conclusion

This study proposes a multimodal deep learning framework for stroke detection that integrates facial landmarks and speech acoustic features using a CNN-GRU architecture. A key contribution, epoch-wise stochastic modality masking, allows the model to learn from facial and auditory data originating from different sources, serving as an effective alternative in scenarios where perfectly aligned multimodal data is not viable or present. Furthermore, the approach enhances robustness by training the model to handle missing or noisy modalities. Given the lightweight nature of the CNN-GRU architecture, the proposed study is well-suited for use in mobile health applications and communities without access to high-end imaging technologies, providing triage support in resource-constrained settings. Furthermore, the framework allows for the democratization of diagnostic tools, empowering community healthcare workers and family caregivers, leading to early detection and better survival rates of stroke victims in low-resource environments. Additionally, the explainable nature of the system ensures that the decisions it gives are not just "black box" output but are interpretable results. This study also provides a detailed analysis of model performance under corrupted and degraded input conditions, simulating realistic scenarios likely to be encountered in practical deployment. The experimental evaluations demonstrate the plausibility of the proposed framework, with the model measuring an accuracy of 98.97% in multimodal settings, as well as 99.66% for audio-only and 87.24% for face-only configurations.

The limitations of this study include several important considerations. First, dataset shift and generalizability. The facial images and audio recordings used in this work are collected from independent repositories, including web-based facial image collections and the TORGO speech database. As such, the data do not qualify as a true multimodal clinical dataset. This cross-source setup limits the model's ability to learn intrinsic modality correspondence that arise due to synchronized visual and auditory manifestations of stroke. Therefore, external validation on prospectively collected clinical multimodal data still remains a critical next step. Second, evaluation strategy. The current study uses a single stratified split of 70/30. In future studies, it is recommended to use nested cross-validation, along with calculating confidence intervals to quantify variance within the modeling. Third, clinical utility and workflow integration. Although the proposed framework shows strong predictive performance, it needs to be ensured that the model is not only accurate but also practical, helping clinicians in the decision-making process, as well as easily fitting into routine hospital processes.

Future work will address these limitations, extending training and validation from larger cohorts with clinical diversity and prospective datasets from hospitals. Cost-sensitive optimizations will be developed to address real-world stroke triage, monitoring drift over long-term deployment, and interfaces of explanation at the subject instance level, to improve usability for clinicians. Adding the multimodal fusion to incorporate more low-cost signals (e.g., gait analysis, vital signs) would also build the resilience of diagnostics in resource-constrained environments. Lastly, optimizing the framework for mobile and edge devices would enable scalable and real-time adoption in telemedicine and prehospital environments.

CRediT authorship contribution statement

Sadman Sakib Alif: Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization. **Md. Yasin Arafat:** Writing – original draft, Visualization, Methodology, Conceptualization. **Md. Mazharul Islam Nibir:** Writing – review & editing, Formal analysis. **Farhana Mehazbin Tusti:** Writing – review & editing, Visualization, Formal analysis. **Aniqua Nusrat Zereen:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The facial image data is available on Kaggle at: https://www.kaggle.com/datasets/thealif/stroke-facial-image-dataset. The speech dataset used for audio analysis is available on Kaggle at: https://www.kaggle.com/datasets/poojag718/dysarthria-and-nondysarthria-speech-dataset.

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